CHAPTER 05

Alcohols

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: eb93a325abc40eeb368706b795a3a69389782001da748f634f79bc1d52e3c7dc

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

ch05:exercise:classify-alcohol | classify-alcohol:stage-1, classify-alcohol:stage-2, classify-alcohol:stage-3

ch05:exercise:oxidize-secondary | oxidize-secondary:stage-1, oxidize-secondary:stage-2, oxidize-secondary:stage-3

ch05:exercise:protection | protection:stage-1, protection:stage-2, protection:stage-3

ch05:exercise:quinone-redox | quinone-redox:stage-1, quinone-redox:stage-2, quinone-redox:stage-3

ch05:lesson:properties | properties:rule, properties:cue, properties:trap, properties:app, properties:step-1, properties:step-2

ch05:lesson:oxidation | oxidation:rule, oxidation:cue, oxidation:trap, oxidation:app, oxidation:step-1, oxidation:step-2, oxidation:step-3

ch05:lesson:activation | activation:rule, activation:cue, activation:trap, activation:app, activation:step-1, activation:step-2, activation:step-3

ch05:lesson:phenols | phenols:rule, phenols:cue, phenols:trap, phenols:app, phenols:step-1, phenols:step-2, phenols:step-3

ch05:molecule:isopropanol | isopropanol

ch05:molecule:tbutanol | tbutanol

ch05:molecule:acetone | acetone

ch05:molecule:acetaldehyde | acetaldehyde

ch05:molecule:mesylate | mesylate

ch05:molecule:acetal | acetal

ch05:molecule:phenol | phenol

ch05:molecule:hydroquinone | hydroquinone

ch05:molecule:quinone | quinone

ch05:curriculum | scope-1, scope-2, scope-3

Curriculum scope

scope-1: 5.1 Description and Properties

scope-2: 5.2 Reactions of Alcohols

scope-3: 5.3 Reactions of Phenols

Objective: Classify alcohols and physical properties

Objective: Predict oxidation and protection outcomes

Objective: Distinguish phenols, quinones, and hydroquinones

Supplied Kaplan curriculum: Chapter 5; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter05b.xhtml

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

properties | Alcohol classes and physical properties

Classify the carbon bearing OH

properties:rule | Core Rules

A primary, secondary, or tertiary alcohol has one, two, or three carbon substituents on the OH-bearing carbon. Hydrogen bonding raises boiling points relative to comparable nonpolar molecules. Small alcohols mix well with water; increasing hydrocarbon size lowers water solubility.

properties:cue | Memory Cues

Count C neighbors of C–OH; do not count the number of OH groups.

properties:trap | Common Mistakes / MCAT Traps

A secondary alcohol is not an alcohol with two hydroxyl groups. Phenol has OH directly attached to an aromatic ring and differs in acidity from an aliphatic alcohol.

properties:app | MCAT Applications

An OH group can increase water affinity and supply a hydrogen-bond donor and acceptor in a binding site, but whole-molecule properties still matter.

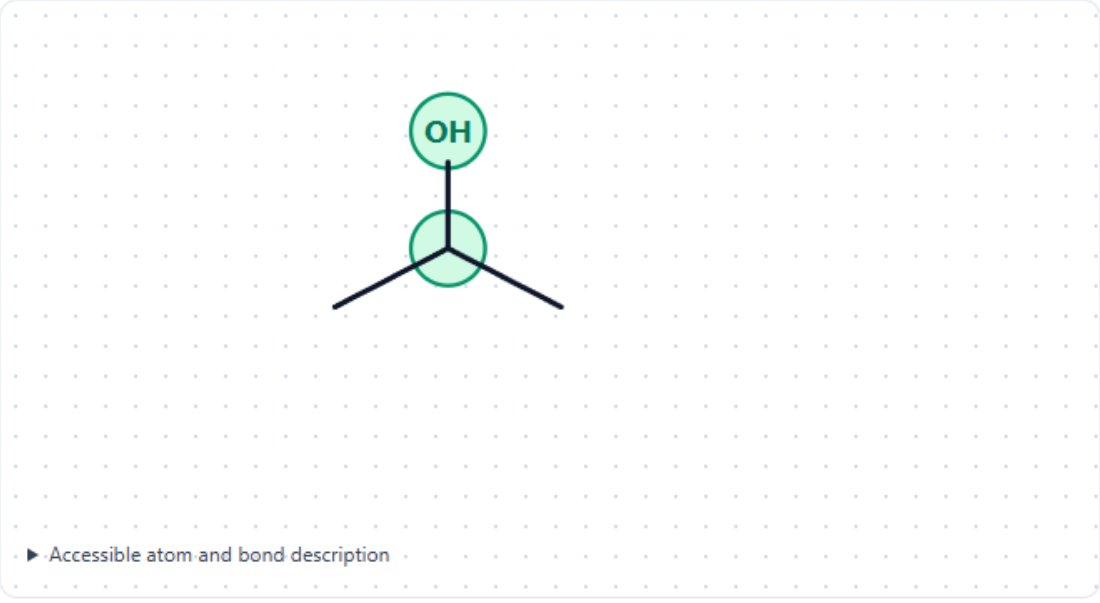
properties:step-1 | 1. Secondary alcohol

The OH-bearing carbon has two carbon neighbors and one H. It is a secondary alcohol.

Displayed molecule ID: isopropanol

{"highlights": {"prin": [1, 3]}, "revealName": true}

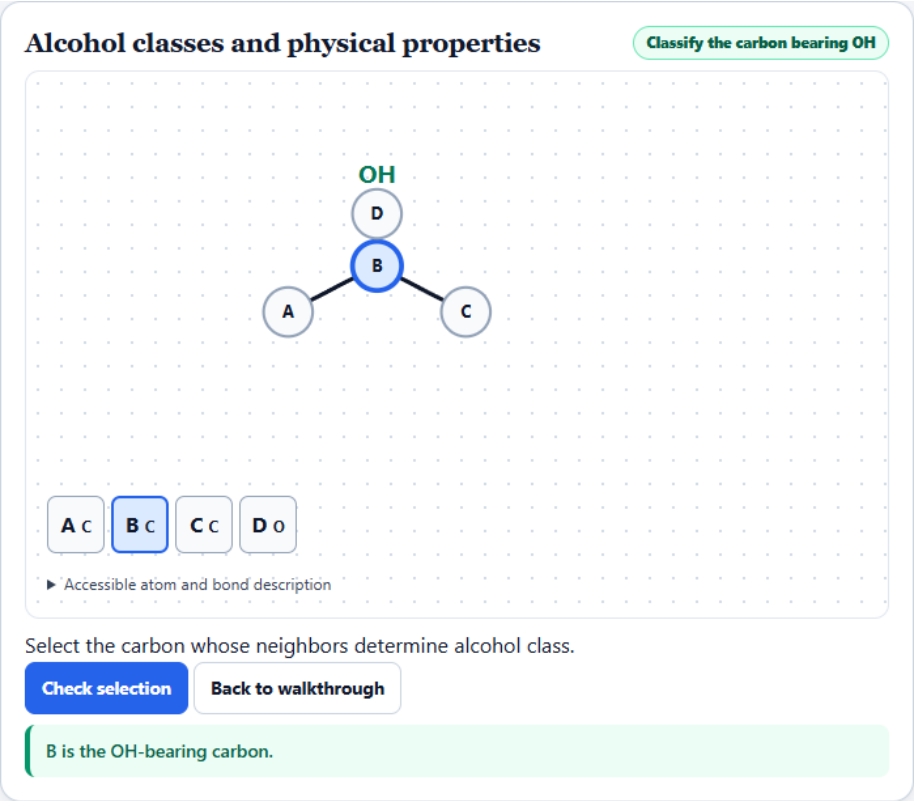
Screenshot: properties-step-1



Embedded challenge: Select the carbon whose neighbors determine alcohol class.

{"prompt": "Select the carbon whose neighbors determine alcohol class.", "answer": [1], "explanation": "B is the OH-bearing carbon."}

Screenshot: properties-step-1-challenge



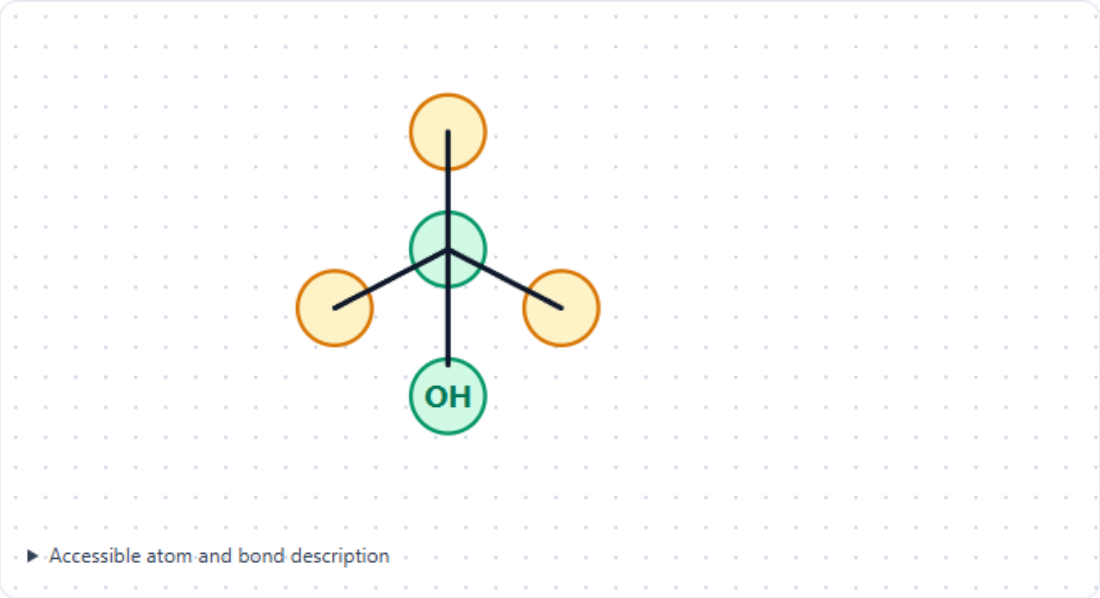
properties:step-2 | 2. Tertiary alcohol

Three carbon groups surround the OH-bearing carbon, which therefore has no C–H bond. This changes its usual oxidation behavior.

Displayed molecule ID: tbutanol

{"highlights": {"prin": [1, 4], "sub": [0, 2, 3]}, "revealName": true}

Screenshot: properties-step-2



oxidation | Predict alcohol oxidation products

Alcohol class + oxidant + conditions

oxidation:rule | Core Rules

Primary alcohols give aldehydes with suitable mild anhydrous oxidation (PCC is one example), or acids with suitable aqueous strong oxidation. Secondary alcohols give ketones. Tertiary alcohols do not undergo the ordinary C–H-removing oxidation without C–C cleavage.

oxidation:cue | Memory Cues

1° → aldehyde → acid; 2° → ketone; 3° lacks the necessary C–H.

oxidation:trap | Common Mistakes / MCAT Traps

PCC is not the only aldehyde-selective oxidant. Do not label every reagent containing oxygen “strong enough” or assume every tertiary alcohol is inert under all conditions.

oxidation:app | MCAT Applications

Alcohol dehydrogenases catalyze related biological redox steps using cofactors; substrate and cofactor specificity control the result.

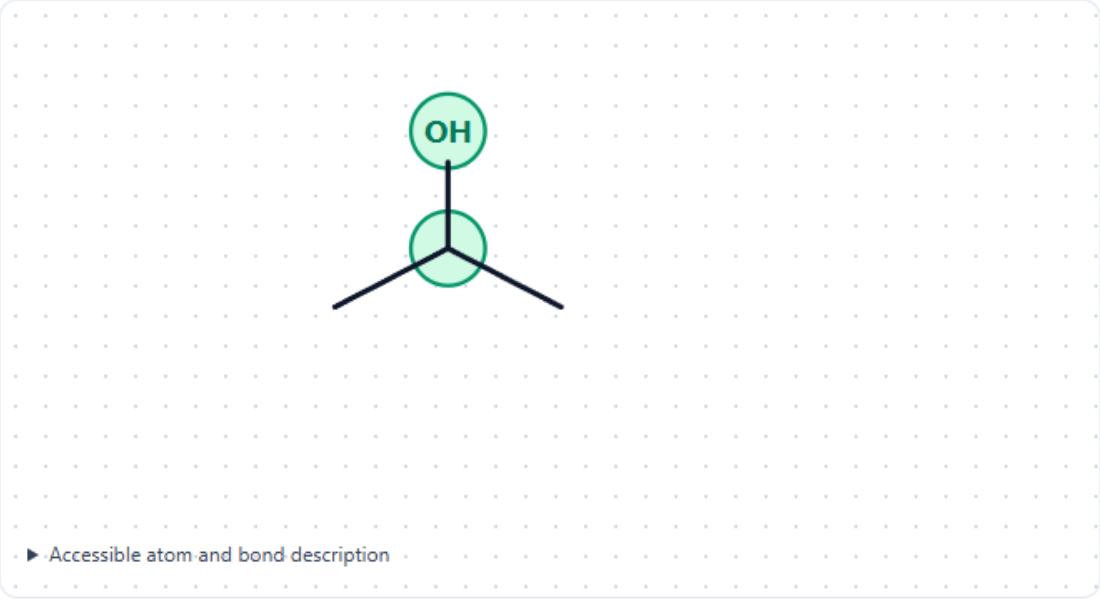
oxidation:step-1 | 1. Locate the oxidized carbon

Oxidizing propan-2-ol removes an O–H hydrogen and the H on its central carbon, producing C=O.

Displayed molecule ID: isopropanol

{"highlights": {"prin": [1, 3]}, "revealName": true}

Screenshot: oxidation-step-1



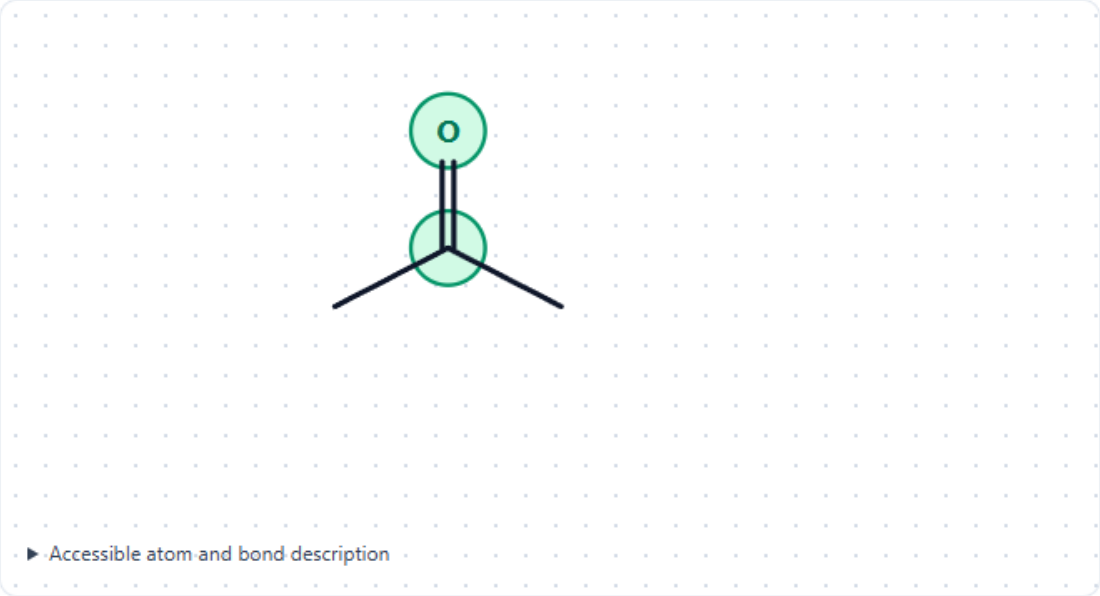
oxidation:step-2 | 2. Compare the product

The same three-carbon framework is now a ketone. Carbon–carbon connectivity is unchanged in this ordinary alcohol oxidation.

Displayed molecule ID: acetone

{"highlights": {"prin": [1, 3]}, "revealName": true}

Screenshot: oxidation-step-2



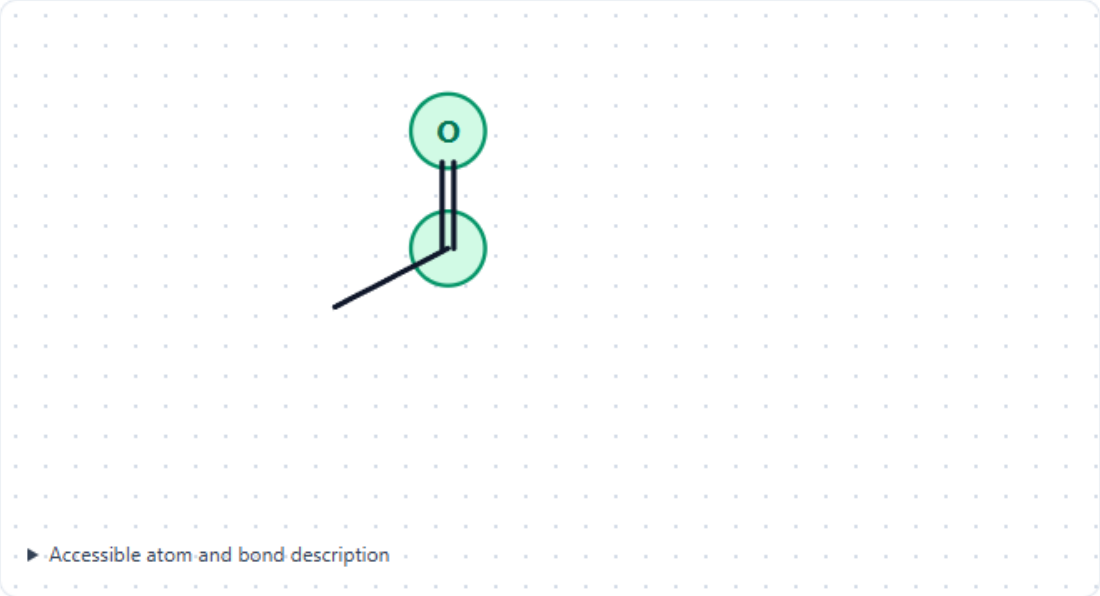
oxidation:step-3 | 3. Primary-alcohol conditions

Ethanal can be obtained from ethanol with an appropriate anhydrous oxidant. In aqueous strongly oxidizing conditions it can be oxidized further to ethanoic acid.

Displayed molecule ID: acetaldehyde

{"highlights": {"prin": [1, 2]}, "revealName": true}

Screenshot: oxidation-step-3



activation | Activate and protect functional groups

Make leaving groups; mask carbonyls

activation:rule | Core Rules

Converting ROH to a mesylate (ROMs) or tosylate (ROTs) gives a much better leaving group. The alcohol oxygen remains bonded to carbon during sulfonate formation. A later SN2 displacement at that carbon inverts its geometry. Acetals protect carbonyls under many basic conditions.

activation:cue | Memory Cues

Activation changes what can leave. Protection temporarily changes what can react.

activation:trap | Common Mistakes / MCAT Traps

Making a tosylate does not itself perform SN2 at carbon. Acetals are generally acid-labile; acid and water are needed for hydrolysis. Protection does not make a group universally inert.

activation:app | MCAT Applications

A carbonyl can be protected while another group reacts, then regenerated. Choose protection and deprotection conditions compatible with the rest of the molecule.

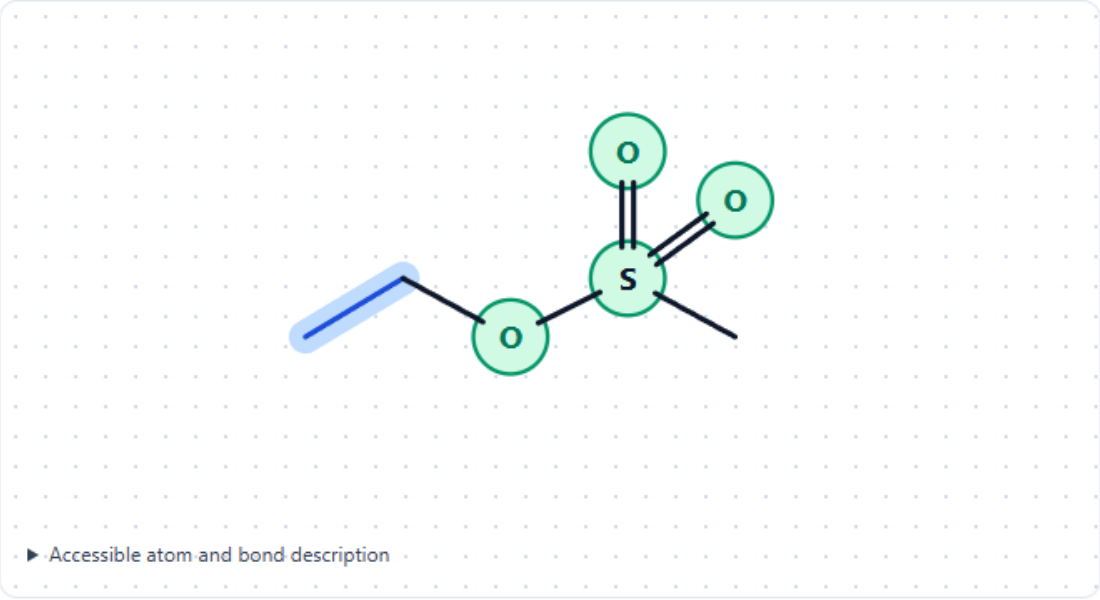
activation:step-1 | 1. Sulfonate ester

The ethyl group retains its C–O bond. The oxygen is now part of a sulfonate ester, whose leaving anion is resonance stabilized.

Displayed molecule ID: mesylate

{"highlights": {"prin": [2, 3, 5, 6], "path": [0, 1]}, "revealName": true}

Screenshot: activation-step-1



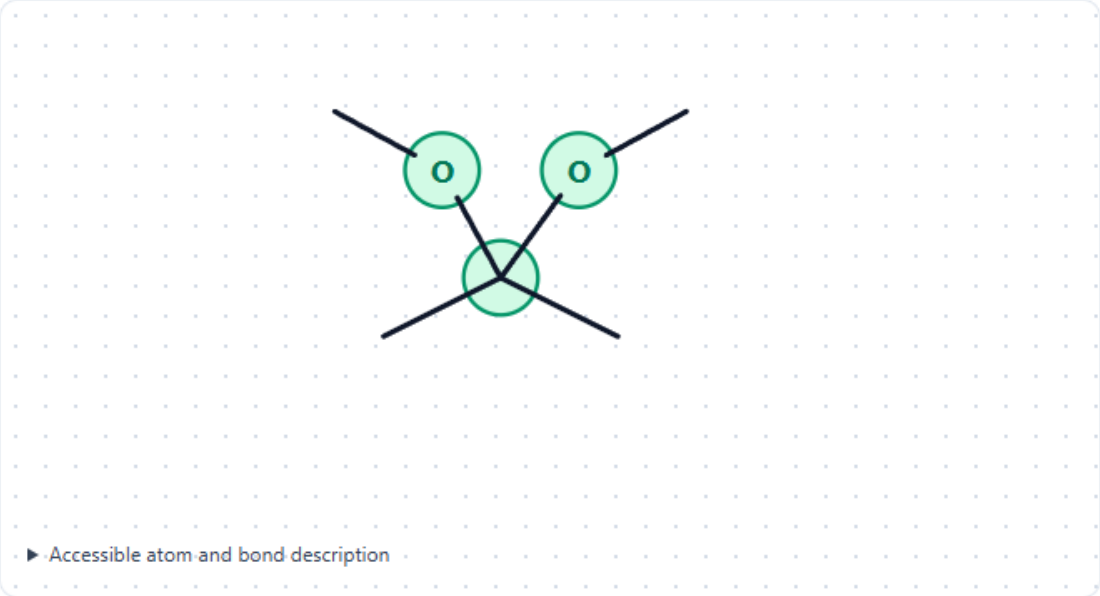
activation:step-2 | 2. Acetal protection

The central carbon has two OR groups and no carbonyl pi bond. Acetals can be made from aldehydes or ketones with alcohol under acid catalysis; removing water favors formation.

Displayed molecule ID: acetal

{"highlights": {"prin": [1, 3, 4]}, "revealName": true}

Screenshot: activation-step-2



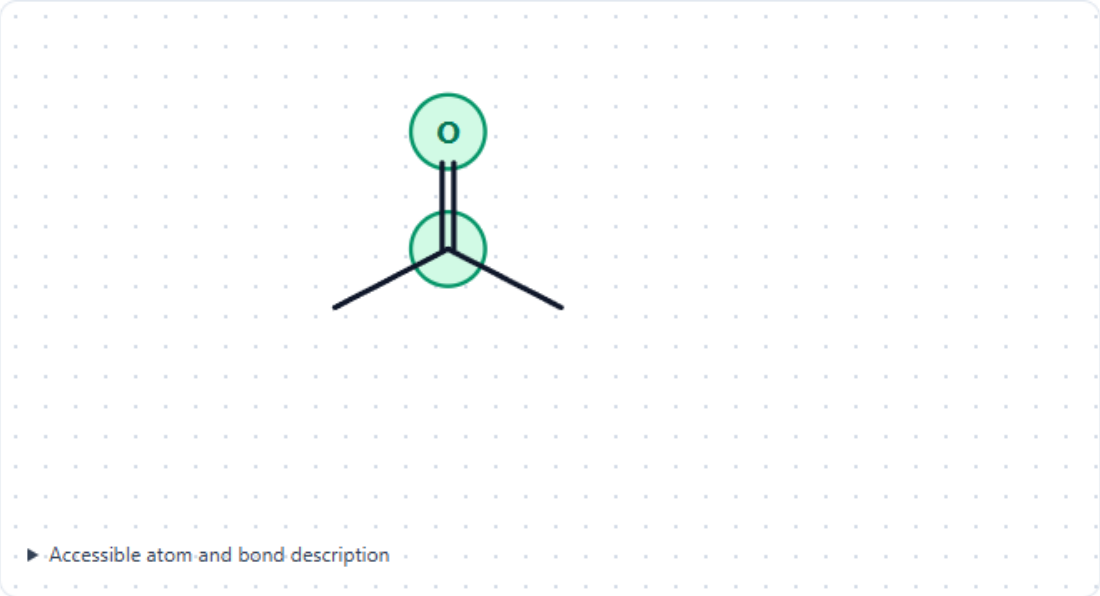
activation:step-3 | 3. Restore the carbonyl

Aqueous acid hydrolyzes the acetal, regenerating the ketone and alcohol. The protection/deprotection sequence is reversible under appropriate conditions.

Displayed molecule ID: acetone

{"highlights": {"prin": [1, 3]}, "revealName": true}

Screenshot: activation-step-3



phenols | Phenols, quinones, and hydroquinones

Acidity and biological electron transfer

phenols:rule | Core Rules

Phenol is more acidic than an ordinary aliphatic alcohol because phenoxide can delocalize negative charge into the ring. Electron-withdrawing substituents can increase acidity. Hydroquinone can be oxidized to a quinone; the reverse is reduction.

phenols:cue | Memory Cues

Quinone + 2 H+ + 2 e− → hydroquinone. “Hydro” here tracks reduction, not merely adding OH groups.

phenols:trap | Common Mistakes / MCAT Traps

Hydroquinone and hydroxyquinone are different terms. Hydroquinone is a reduced quinone form; a hydroxyquinone still has a quinone framework bearing an extra OH. Coenzyme Q accepts electrons upstream of complex III and delivers them to complex III.

phenols:app | MCAT Applications

Ubiquinone/ubiquinol transports electrons within the mitochondrial inner membrane; proton-coupled redox chemistry links the organic structure to energy metabolism.

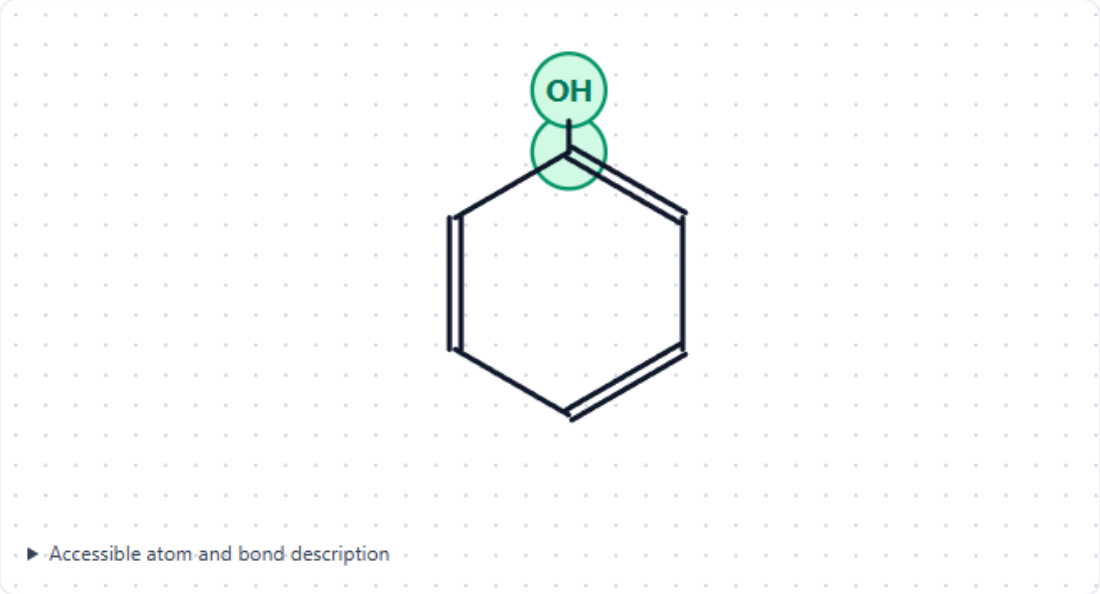
phenols:step-1 | 1. Phenolic OH

OH is attached directly to an aromatic carbon. Deprotonation gives phenoxide, stabilized by resonance. The pictured alternating bonds are one aromatic resonance representation.

Displayed molecule ID: phenol

{"highlights": {"prin": [0, 6]}, "revealName": true}

Screenshot: phenols-step-1



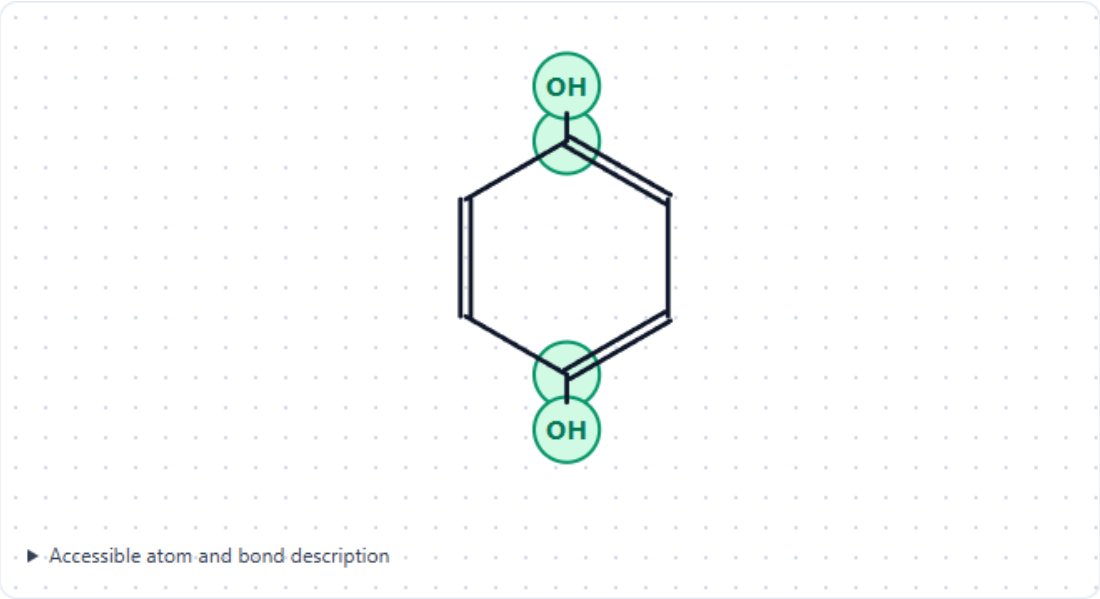
phenols:step-2 | 2. Reduced member

Hydroquinone has two phenolic OH groups in a para arrangement. Oxidation removes two H atoms overall, formally two protons plus two electrons.

Displayed molecule ID: hydroquinone

{"highlights": {"prin": [0, 3, 6, 7]}, "revealName": true}

Screenshot: phenols-step-2



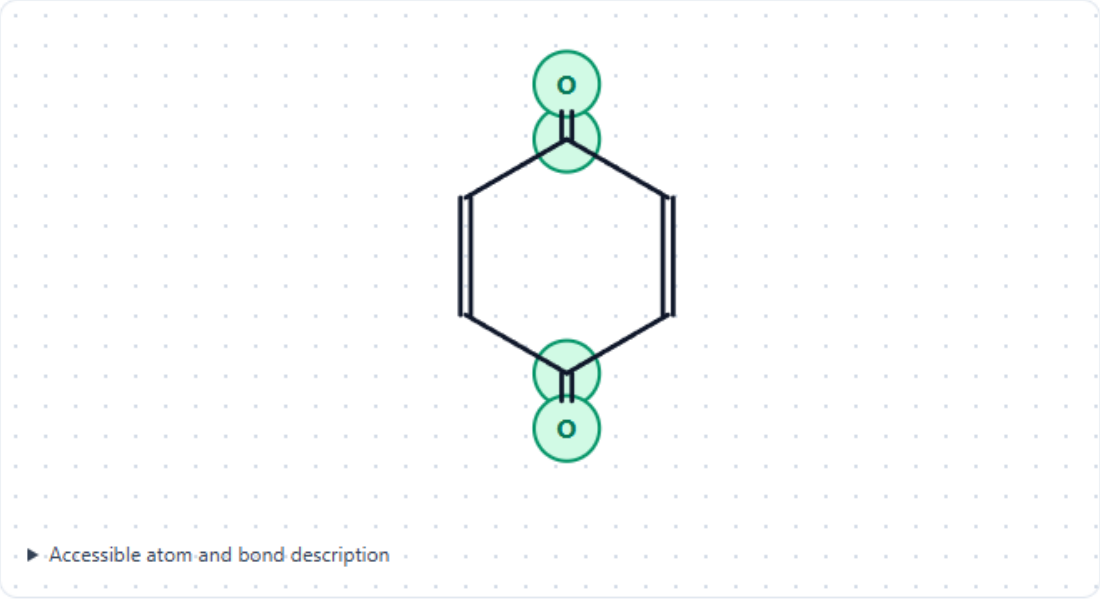
phenols:step-3 | 3. Oxidized member

The quinone has two carbonyls. Its ring is not simply benzene with unchanged aromatic bonding; the pi-bond arrangement has changed.

Displayed molecule ID: quinone

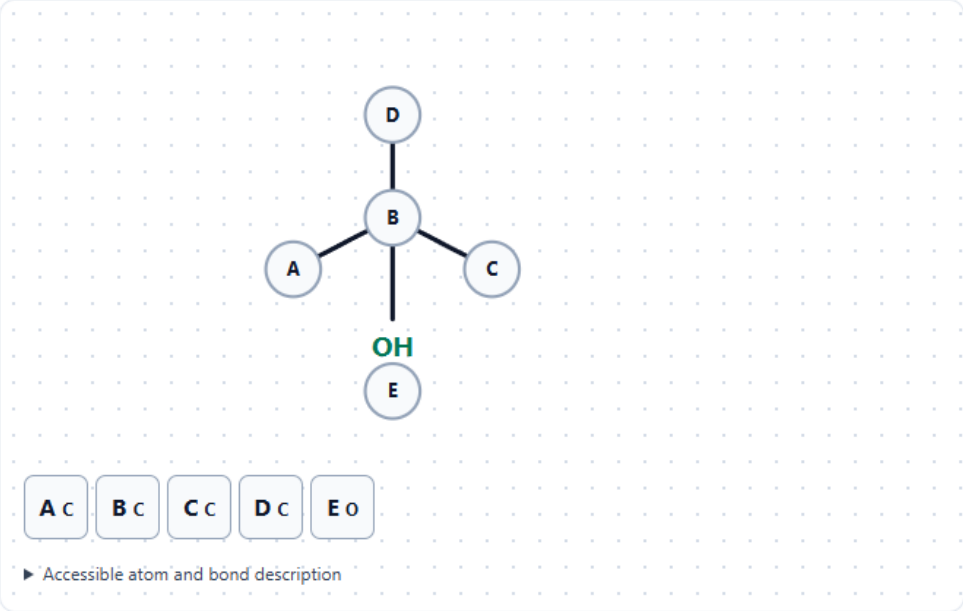
{"highlights": {"prin": [0, 3, 6, 7]}, "revealName": true}

Screenshot: phenols-step-3



classify-alcohol | Classify by carbon neighbors

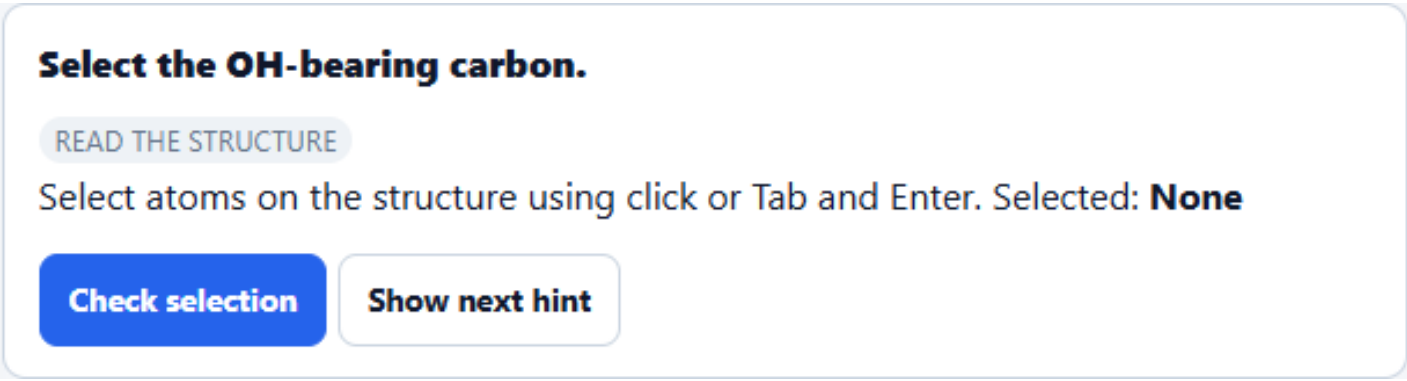
Screenshot: classify-alcohol-stage-1-molecule



classify-alcohol:stage-1 | Select the OH-bearing carbon.

Answer type: atoms

Screenshot: classify-alcohol-stage-1-question



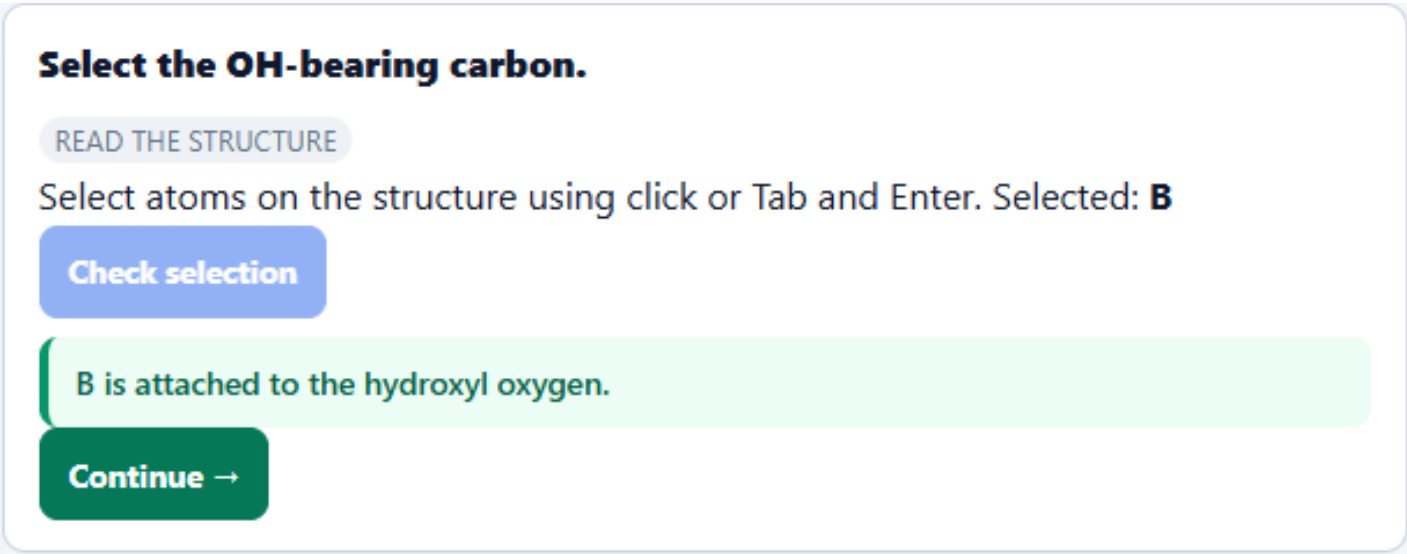
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 4]}}

Hint 1: Find the C–O bond.

Hint 2: Choose the carbon, not the oxygen.

Explanation: B is attached to the hydroxyl oxygen.

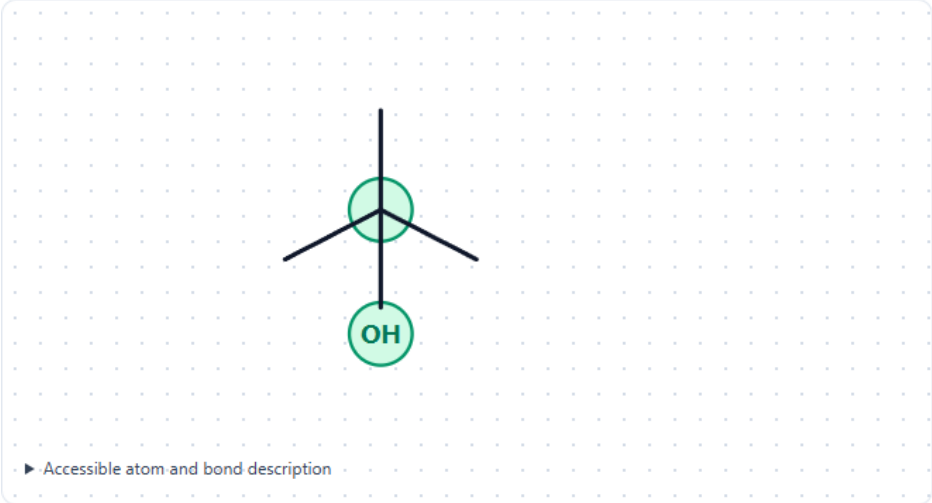
Screenshot: classify-alcohol-stage-1-feedback



classify-alcohol:stage-2 | How many carbon neighbors does it have?

Answer type: number | unit: carbon neighbors

Screenshot: classify-alcohol-stage-2-molecule



Screenshot: classify-alcohol-stage-2-question



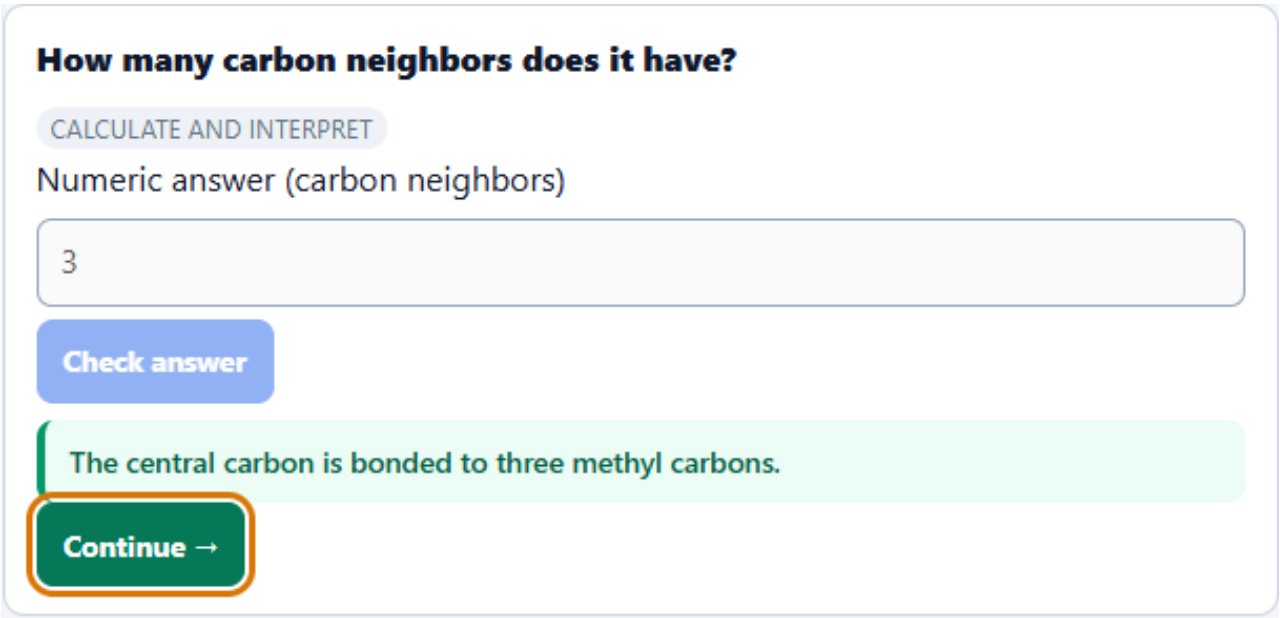
{"authorAnswer": 3, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Count only directly attached C atoms.

Hint 2: Exclude oxygen and implicit H.

Explanation: The central carbon is bonded to three methyl carbons.

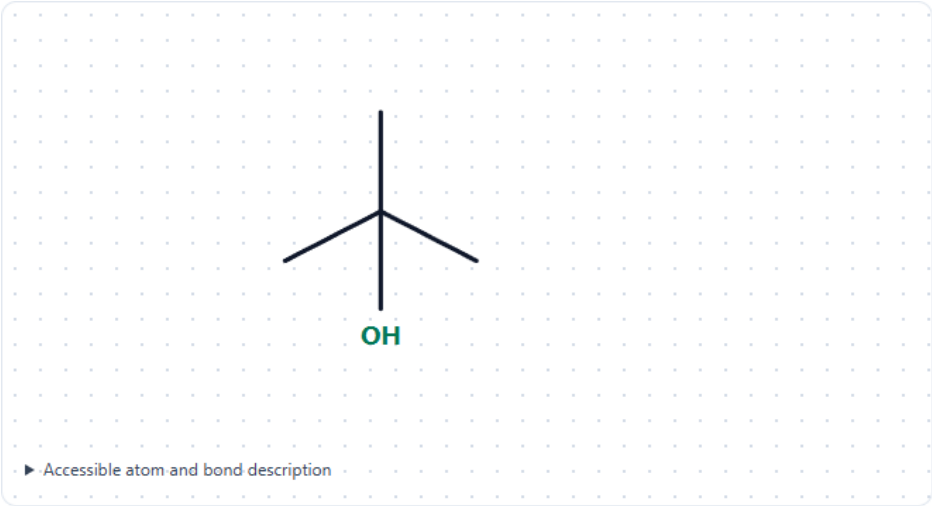
Screenshot: classify-alcohol-stage-2-feedback



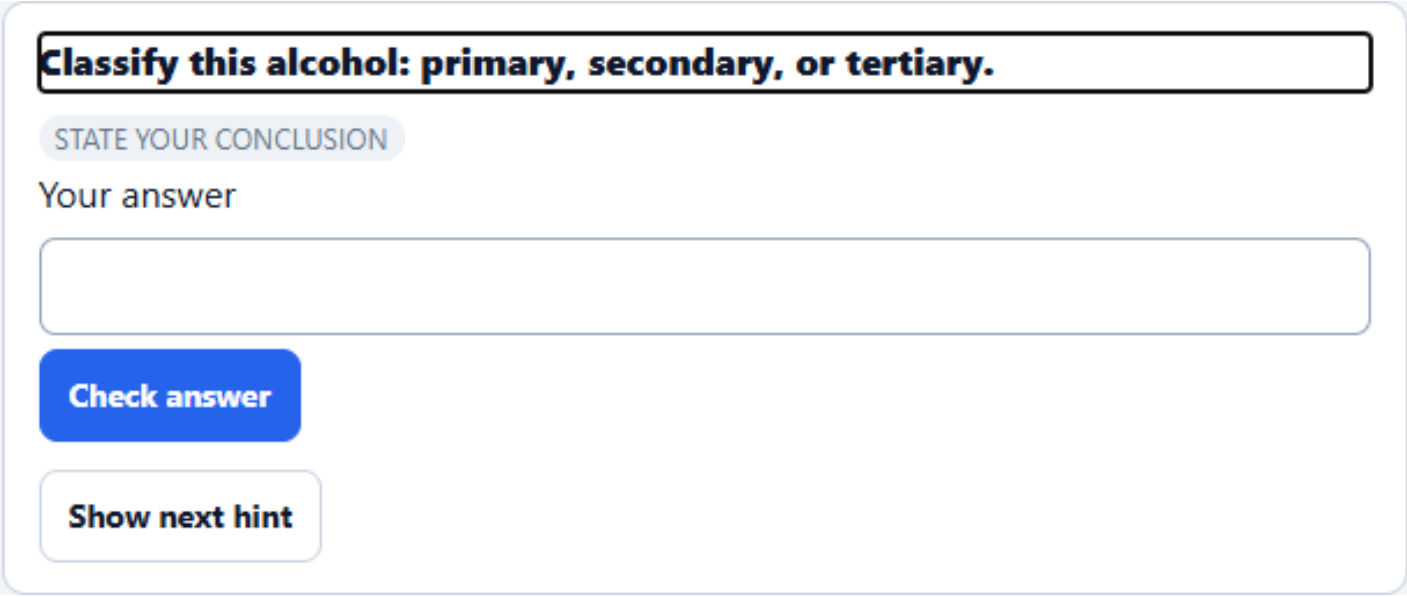
classify-alcohol:stage-3 | Classify this alcohol: primary, secondary, or tertiary.

Answer type: text

Screenshot: classify-alcohol-stage-3-molecule



Screenshot: classify-alcohol-stage-3-question



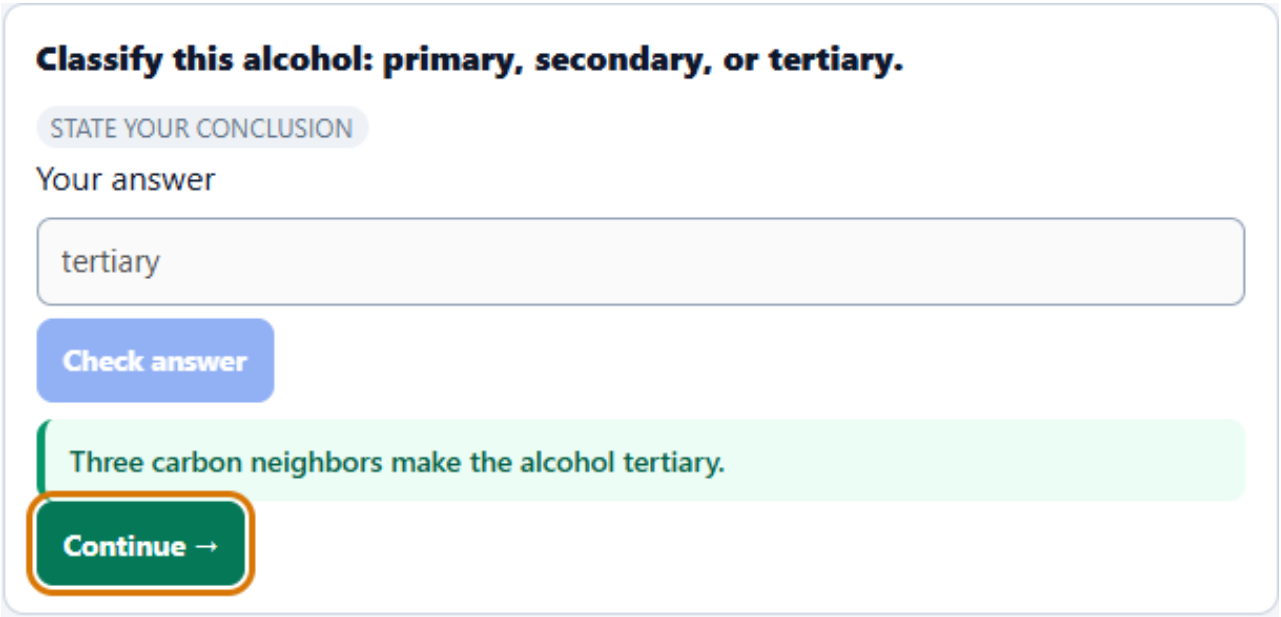
{"authorAnswer": ["tertiary", "3°", "3"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use the carbon-neighbor count.

Hint 2: The OH-bearing carbon has no hydrogen.

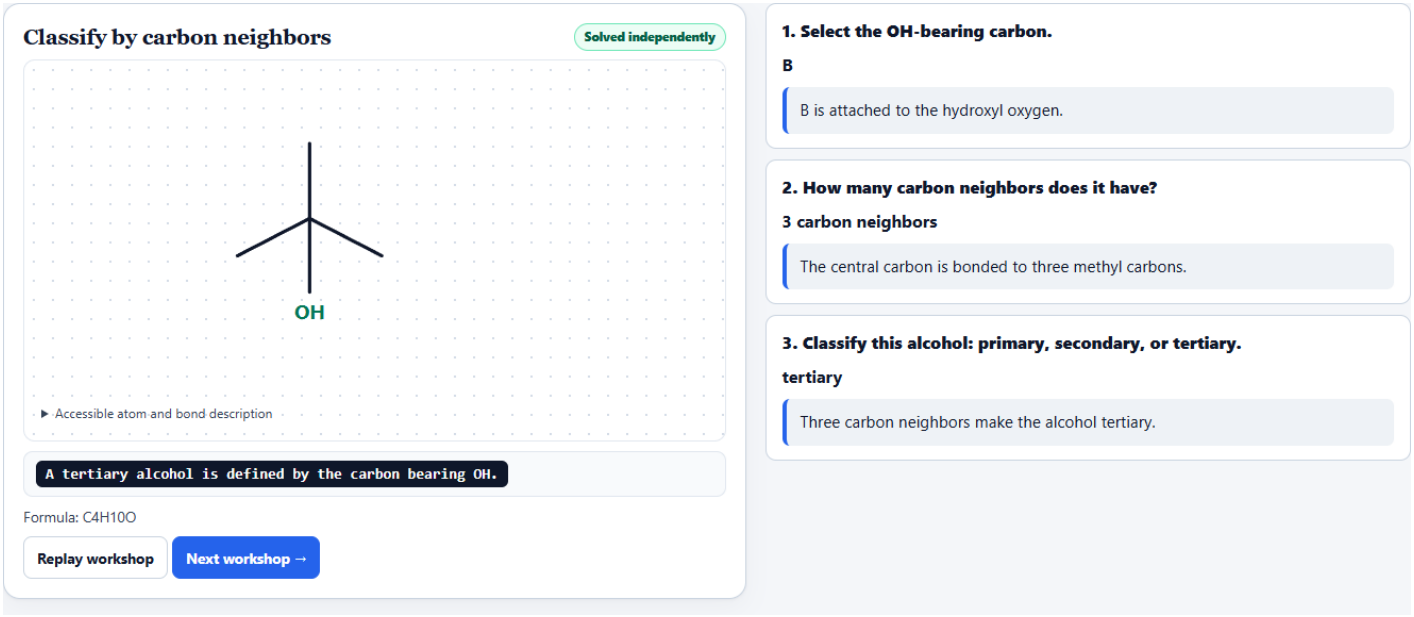
Explanation: Three carbon neighbors make the alcohol tertiary.

Screenshot: classify-alcohol-stage-3-feedback



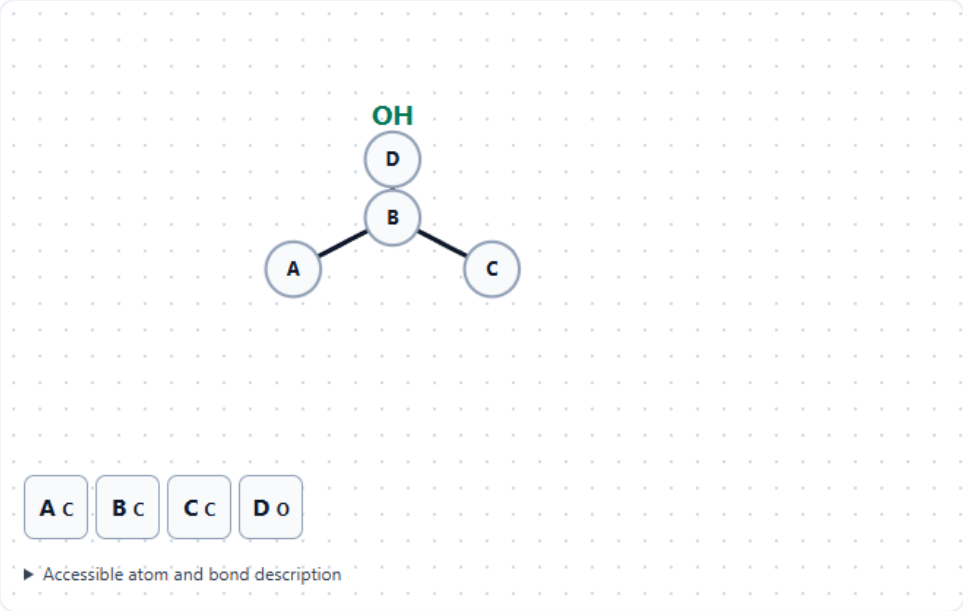
Completion summary: A tertiary alcohol is defined by the carbon bearing OH.

Screenshot: classify-alcohol-complete



oxidize-secondary | Predict the oxidation product

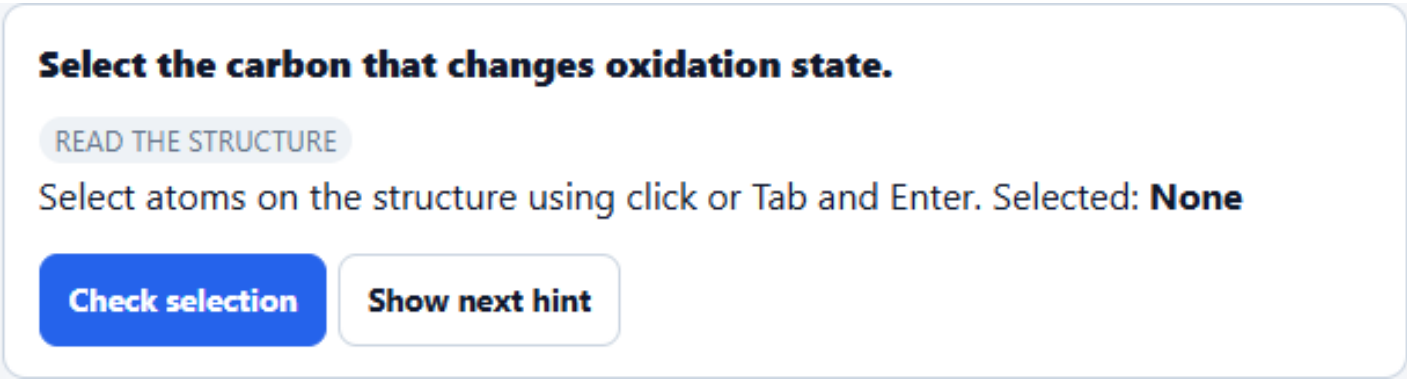
Screenshot: oxidize-secondary-stage-1-molecule



oxidize-secondary:stage-1 | Select the carbon that changes oxidation state.

Answer type: atoms

Screenshot: oxidize-secondary-stage-1-question



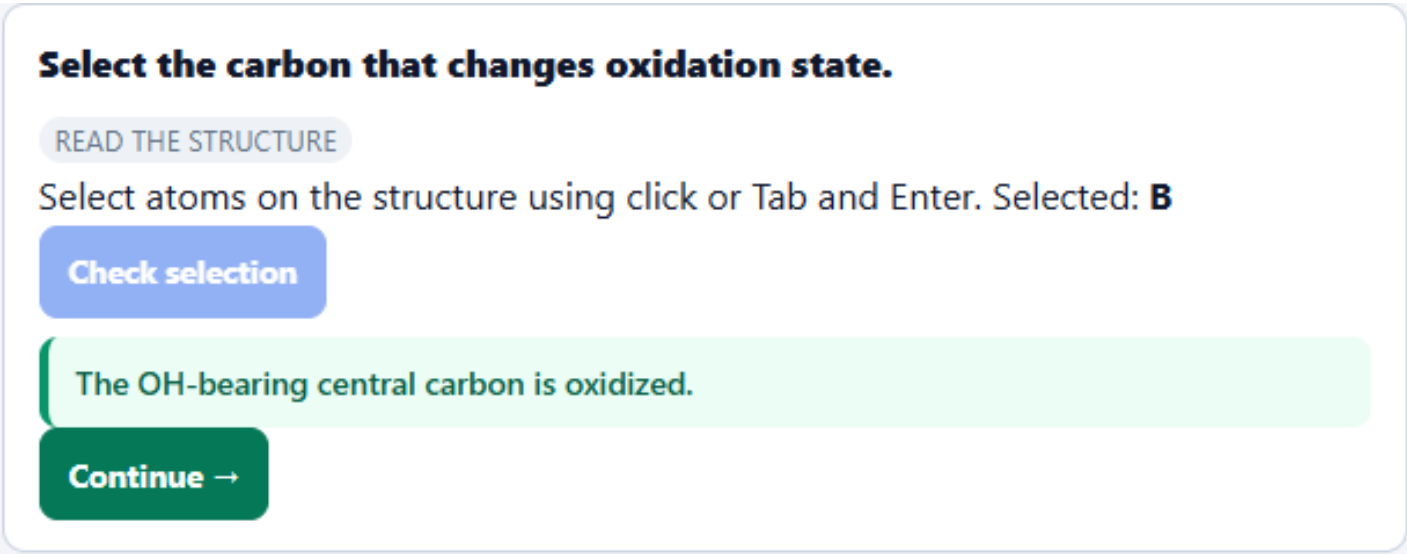
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 3]}}

Hint 1: The carbon skeleton stays intact.

Hint 2: Track the C–O bond.

Explanation: The OH-bearing central carbon is oxidized.

Screenshot: oxidize-secondary-stage-1-feedback



oxidize-secondary:stage-2 | Ordinary oxidation produces which functional group?

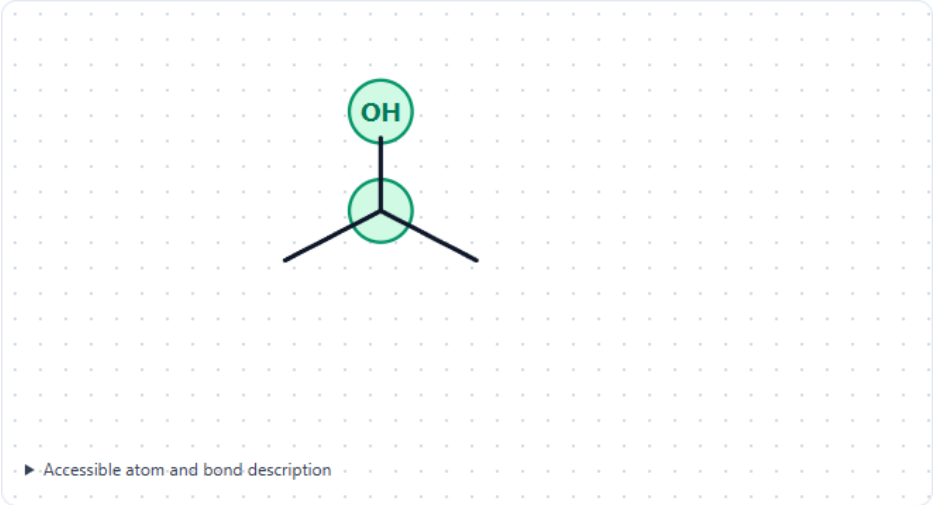
Answer type: choice

Option 0 (A): Ketone

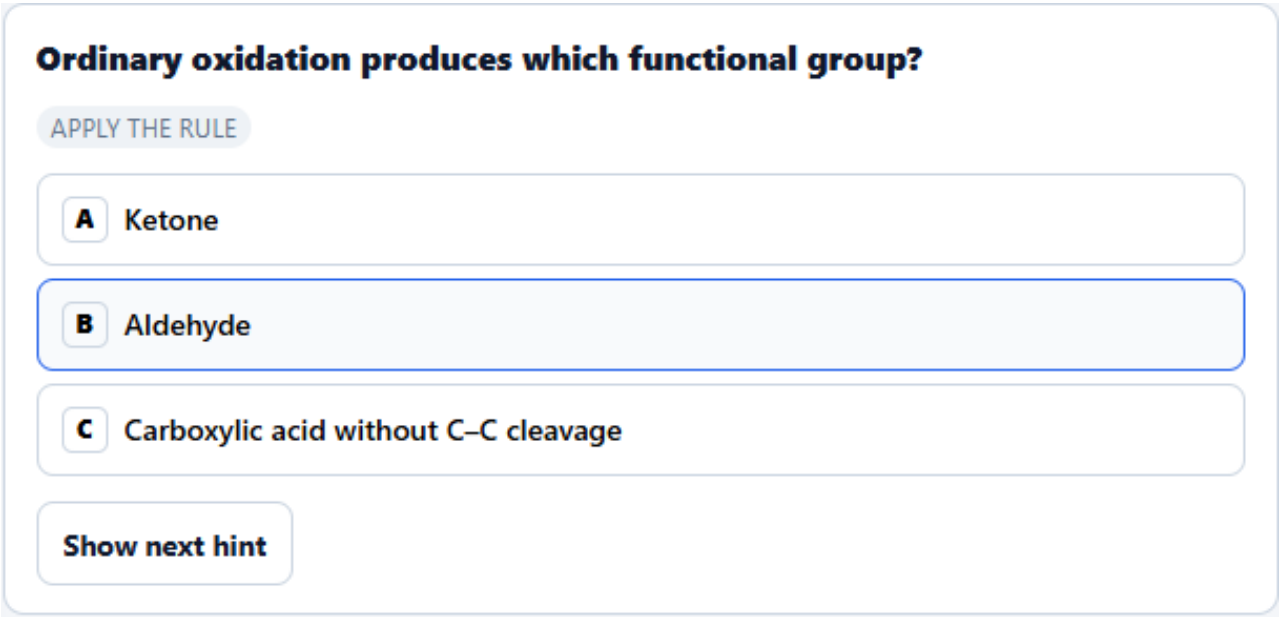
Option 1 (B): Aldehyde

Option 2 (C): Carboxylic acid without C–C cleavage

Screenshot: oxidize-secondary-stage-2-molecule



Screenshot: oxidize-secondary-stage-2-question



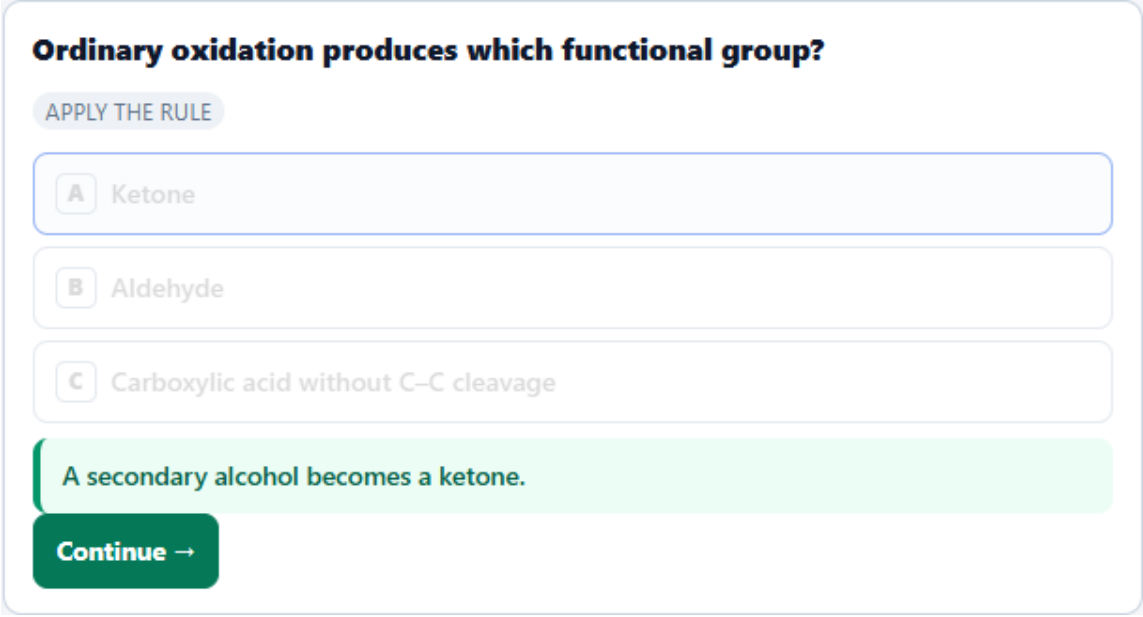
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: How many carbon substituents remain on the carbonyl?

Hint 2: Two carbon neighbors indicate a ketone.

Explanation: A secondary alcohol becomes a ketone.

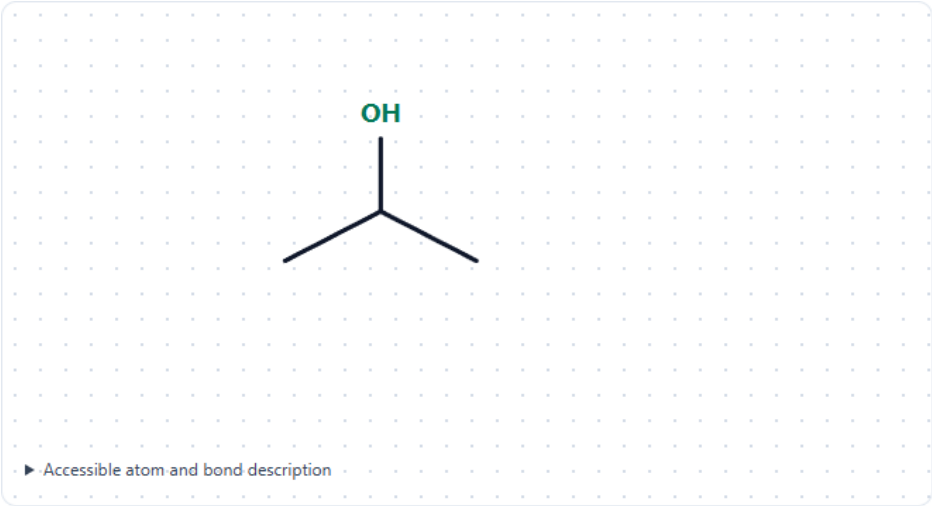
Screenshot: oxidize-secondary-stage-2-feedback



oxidize-secondary:stage-3 | Name the three-carbon product.

Answer type: text

Screenshot: oxidize-secondary-stage-3-molecule



Screenshot: oxidize-secondary-stage-3-question



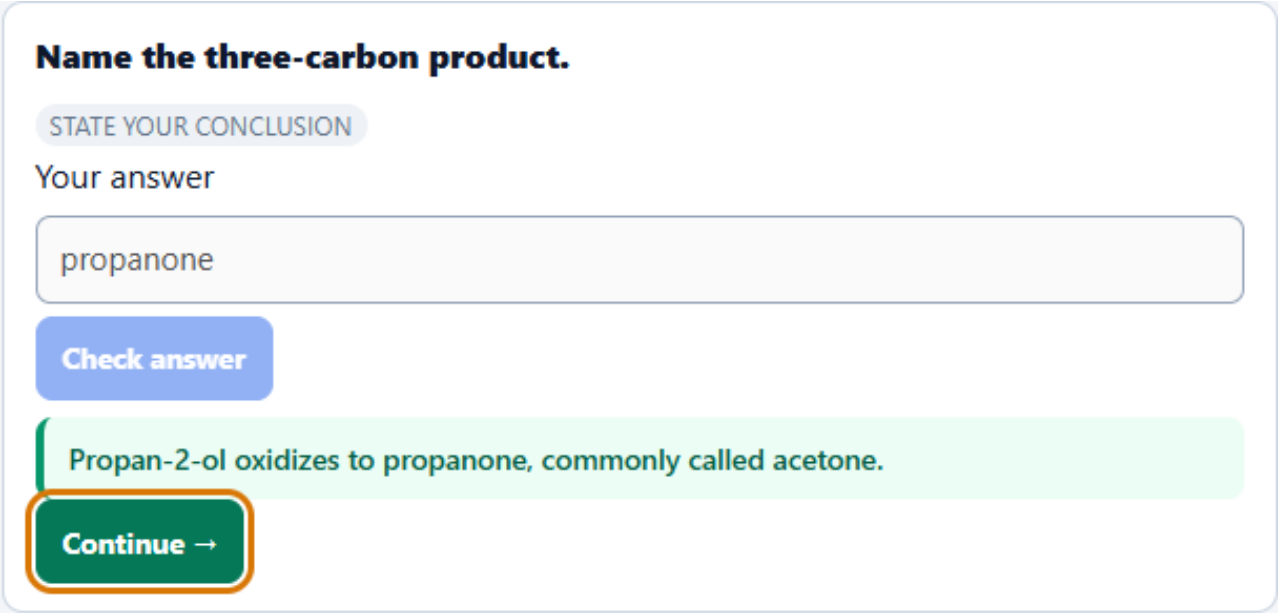
{"authorAnswer": ["propanone", "propan-2-one", "acetone"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Retain all three carbons.

Hint 2: Replace the secondary alcohol with a ketone.

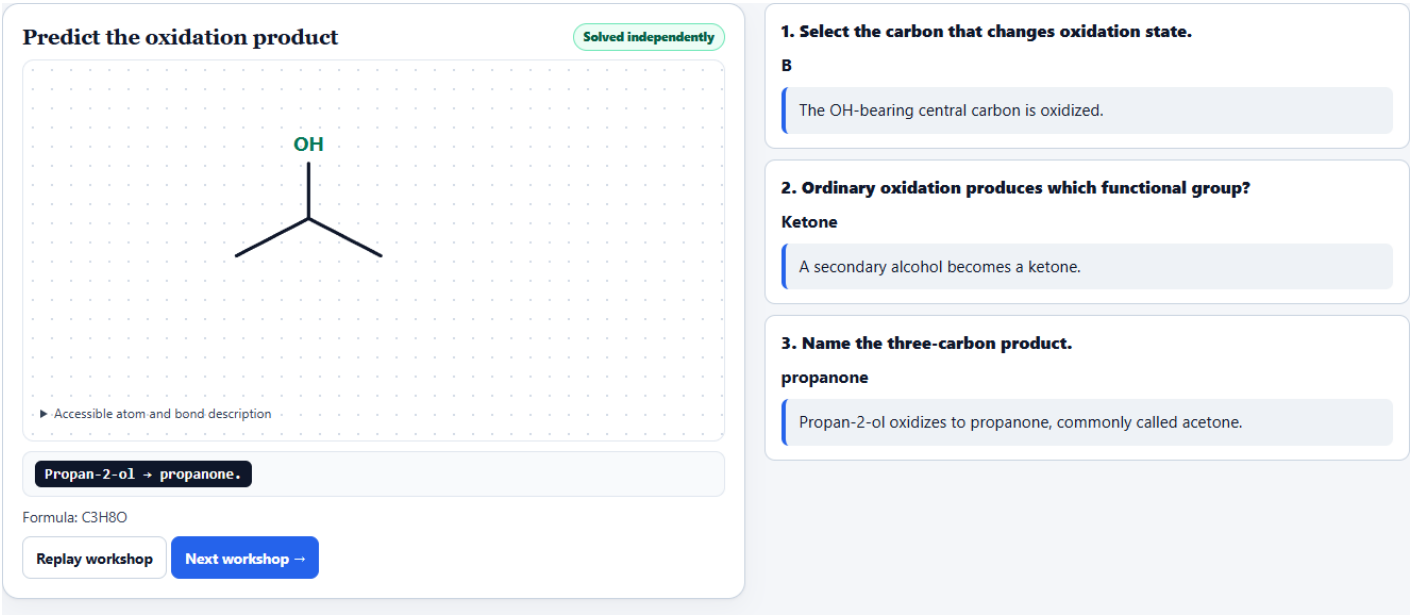
Explanation: Propan-2-ol oxidizes to propanone, commonly called acetone.

Screenshot: oxidize-secondary-stage-3-feedback



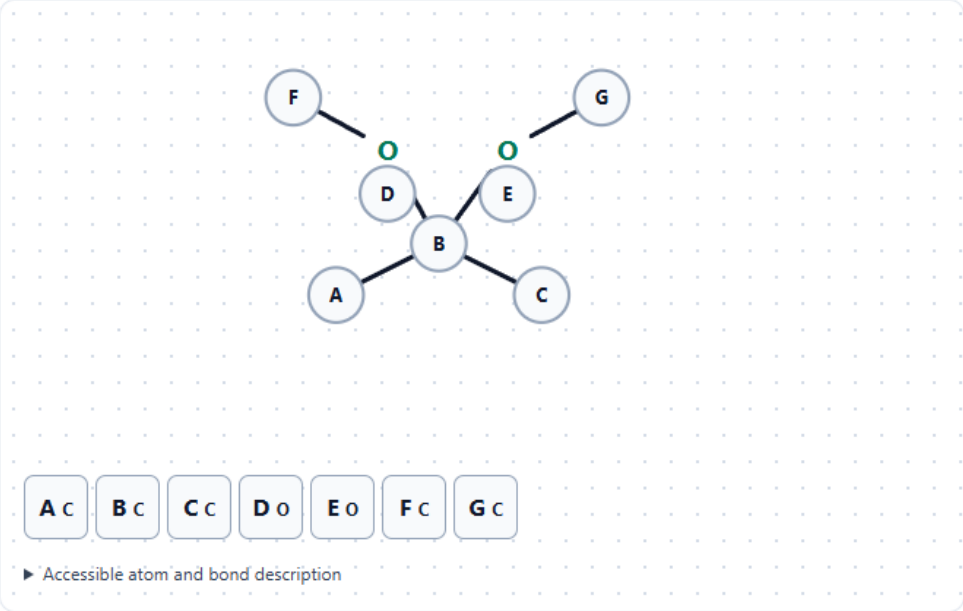
Completion summary: Propan-2-ol → propanone.

Screenshot: oxidize-secondary-complete



protection | Use a protecting group correctly

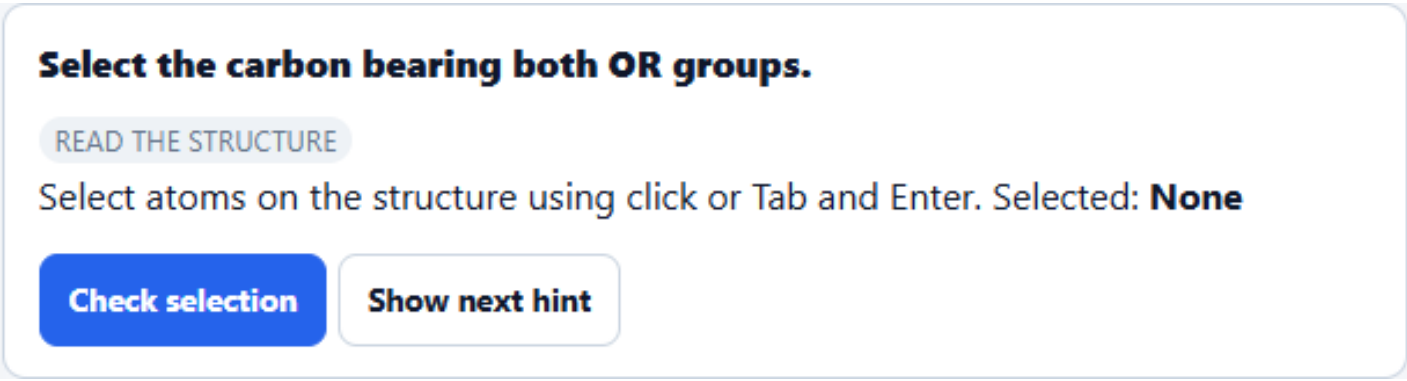
Screenshot: protection-stage-1-molecule



protection:stage-1 | Select the carbon bearing both OR groups.

Answer type: atoms

Screenshot: protection-stage-1-question



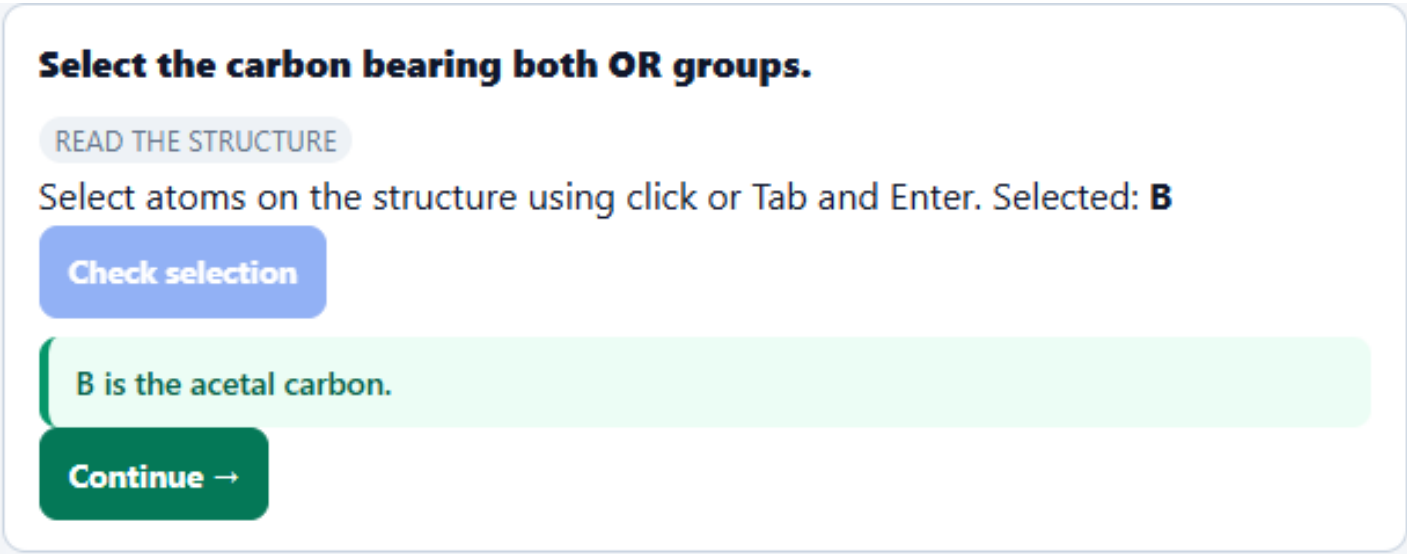
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 3, 4]}}

Hint 1: Find a carbon attached to two oxygens.

Hint 2: There is no C=O at that carbon.

Explanation: B is the acetal carbon.

Screenshot: protection-stage-1-feedback



protection:stage-2 | Which conditions ordinarily regenerate the ketone?

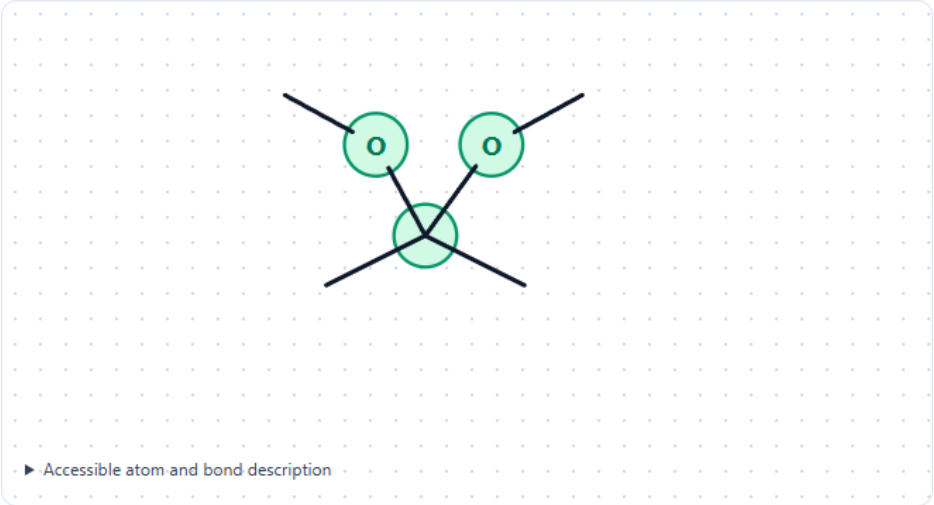
Answer type: choice

Option 0 (A): Aqueous acid

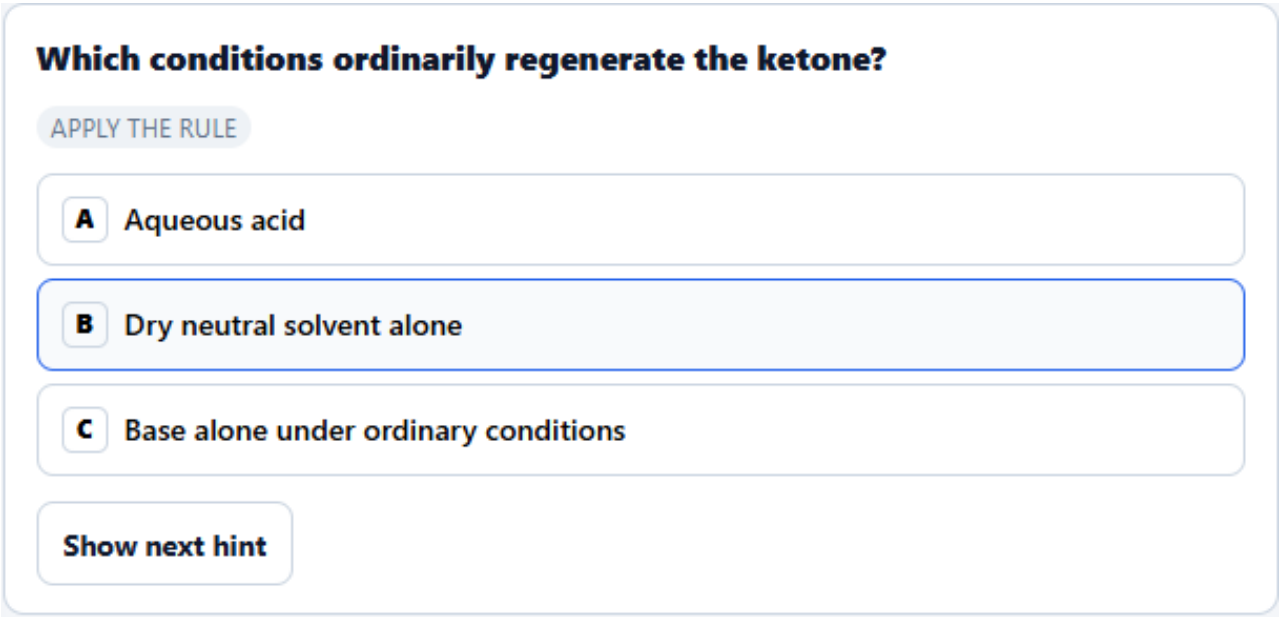
Option 1 (B): Dry neutral solvent alone

Option 2 (C): Base alone under ordinary conditions

Screenshot: protection-stage-2-molecule



Screenshot: protection-stage-2-question



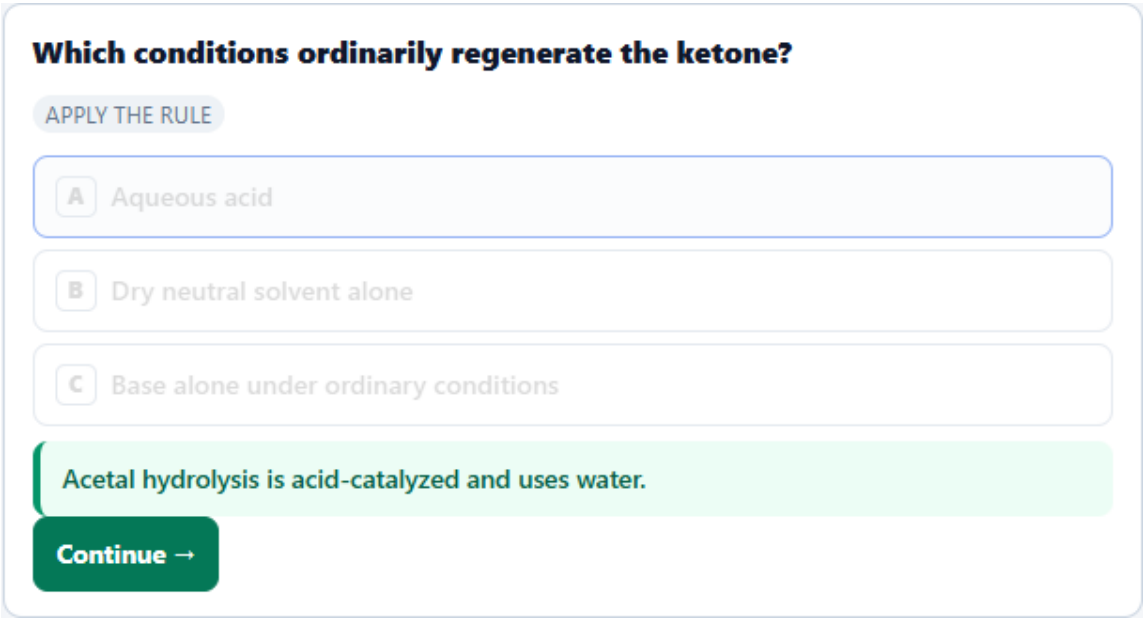
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Consider the reverse of acetal formation.

Hint 2: Both catalyst and water are important.

Explanation: Acetal hydrolysis is acid-catalyzed and uses water.

Screenshot: protection-stage-2-feedback



protection:stage-3 | Making an alkyl tosylate from ROH directly causes…

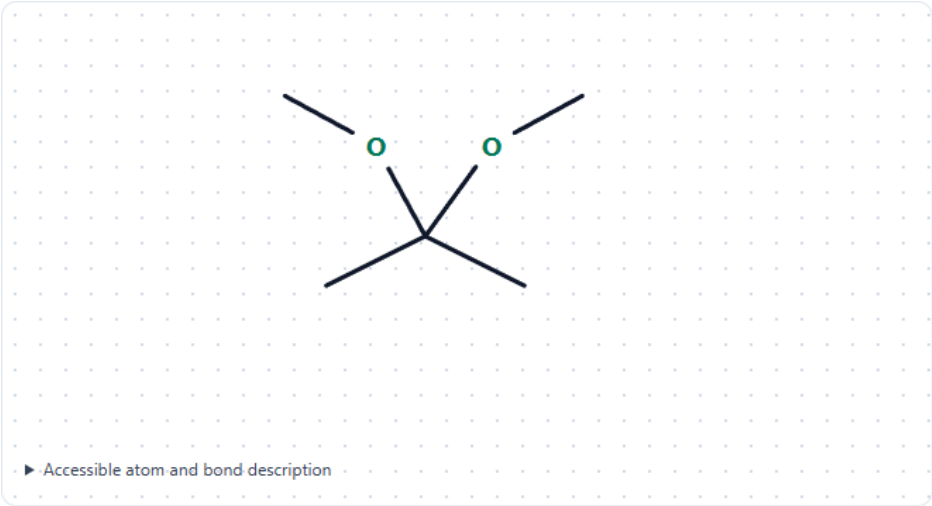
Answer type: choice

Option 0 (A): Retention of the C–O framework

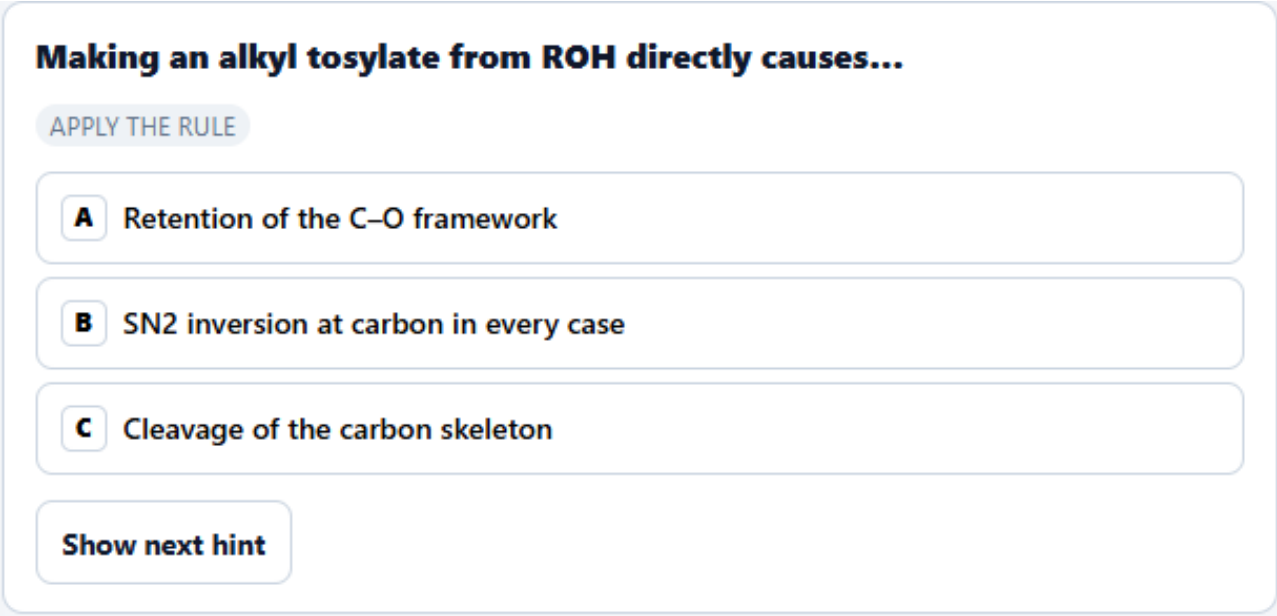
Option 1 (B): SN2 inversion at carbon in every case

Option 2 (C): Cleavage of the carbon skeleton

Screenshot: protection-stage-3-molecule



Screenshot: protection-stage-3-question



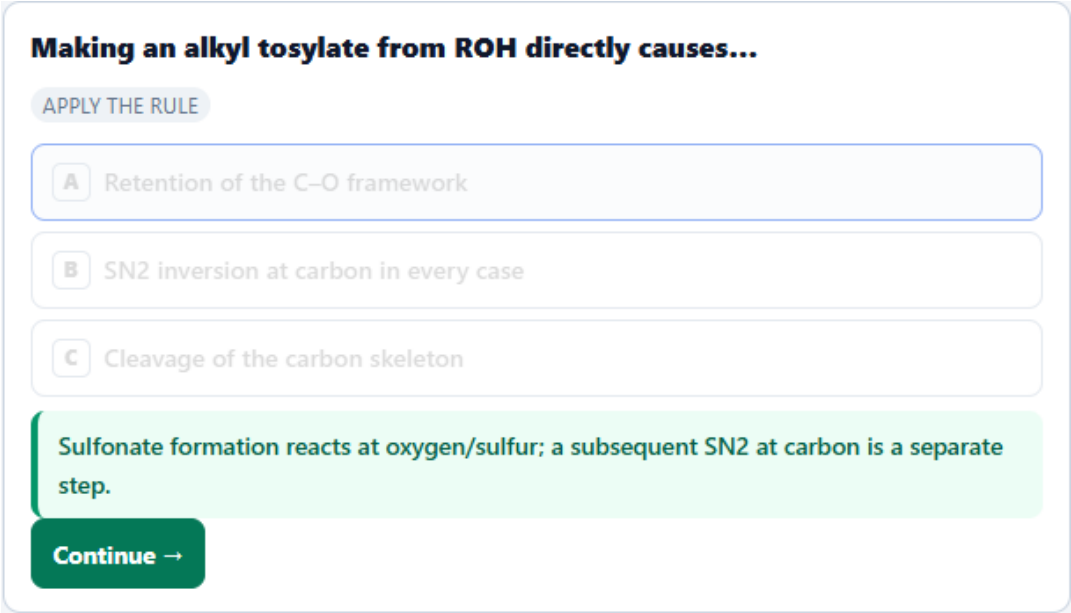
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Distinguish activation from displacement.

Hint 2: The original C–O bond remains during activation.

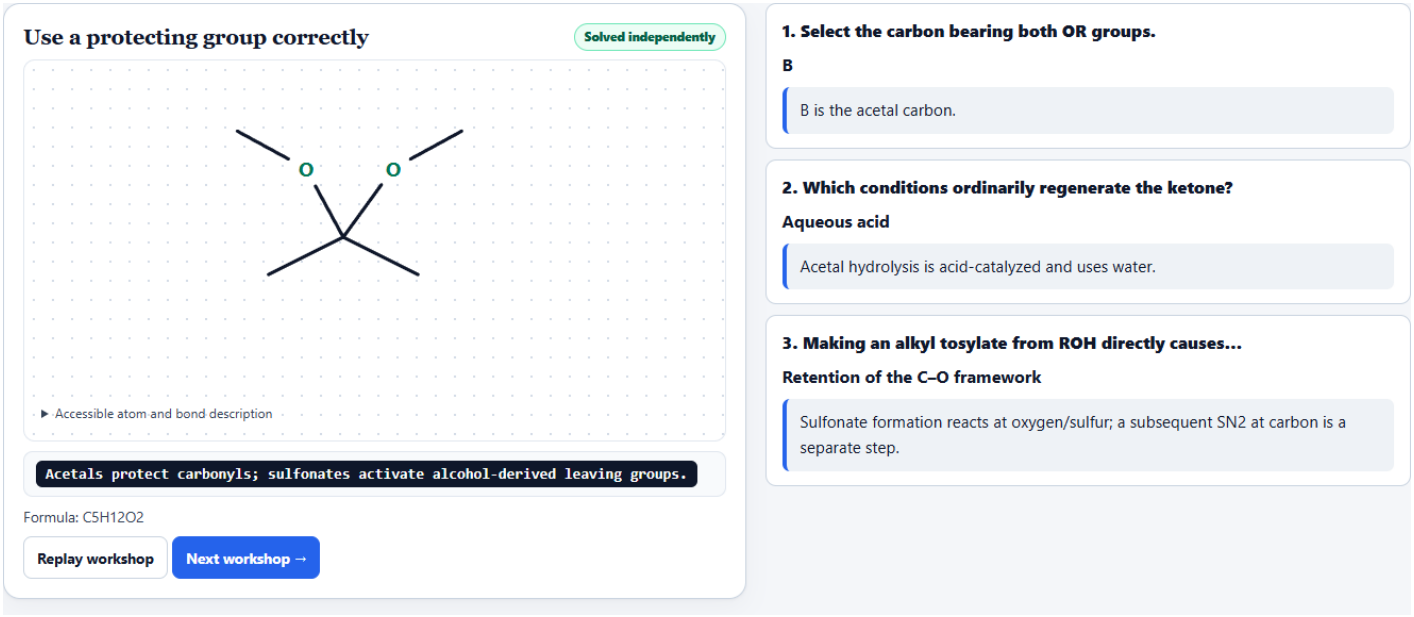
Explanation: Sulfonate formation reacts at oxygen/sulfur; a subsequent SN2 at carbon is a separate step.

Screenshot: protection-stage-3-feedback



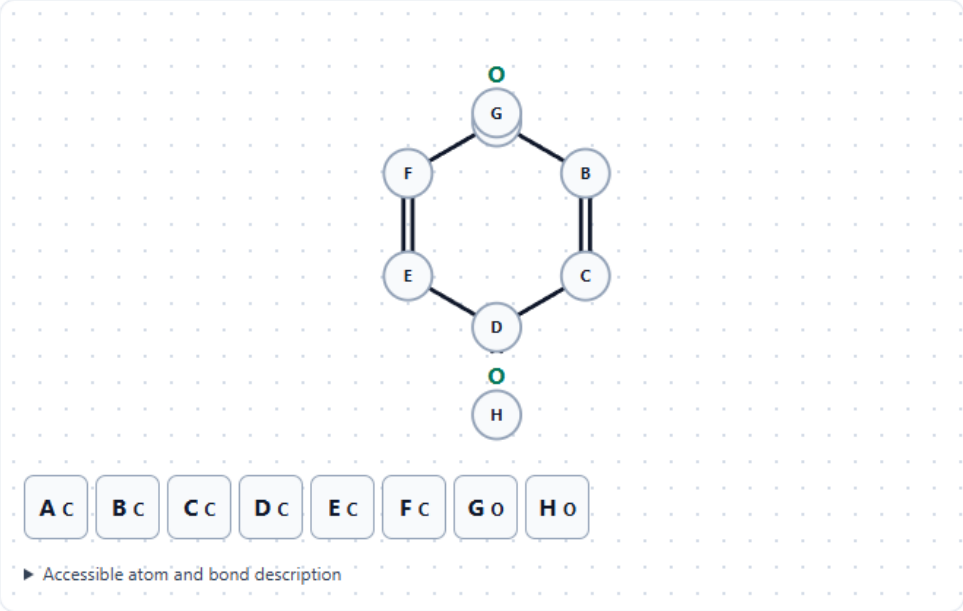
Completion summary: Acetals protect carbonyls; sulfonates activate alcohol-derived leaving groups.

Screenshot: protection-complete



quinone-redox | Track quinone reduction

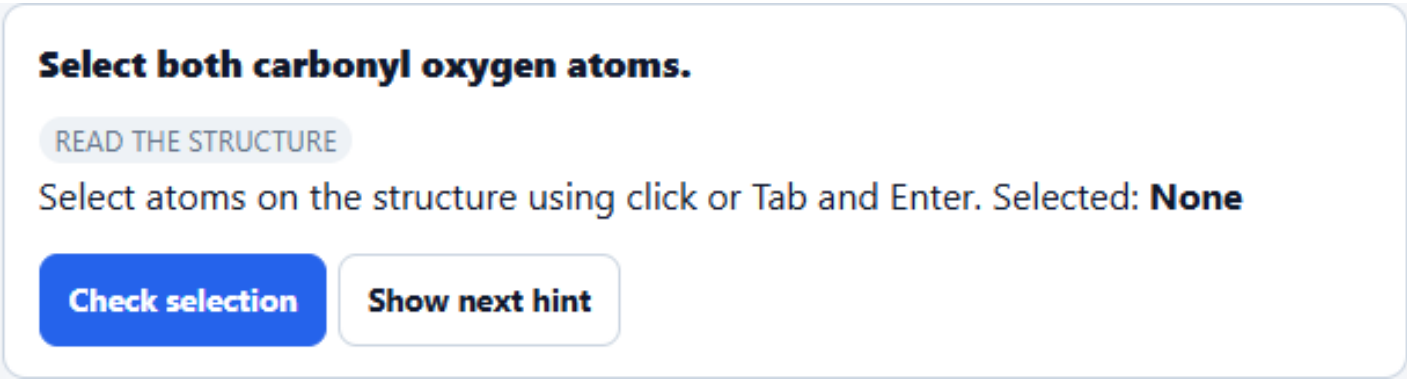
Screenshot: quinone-redox-stage-1-molecule



quinone-redox:stage-1 | Select both carbonyl oxygen atoms.

Answer type: atoms

Screenshot: quinone-redox-stage-1-question



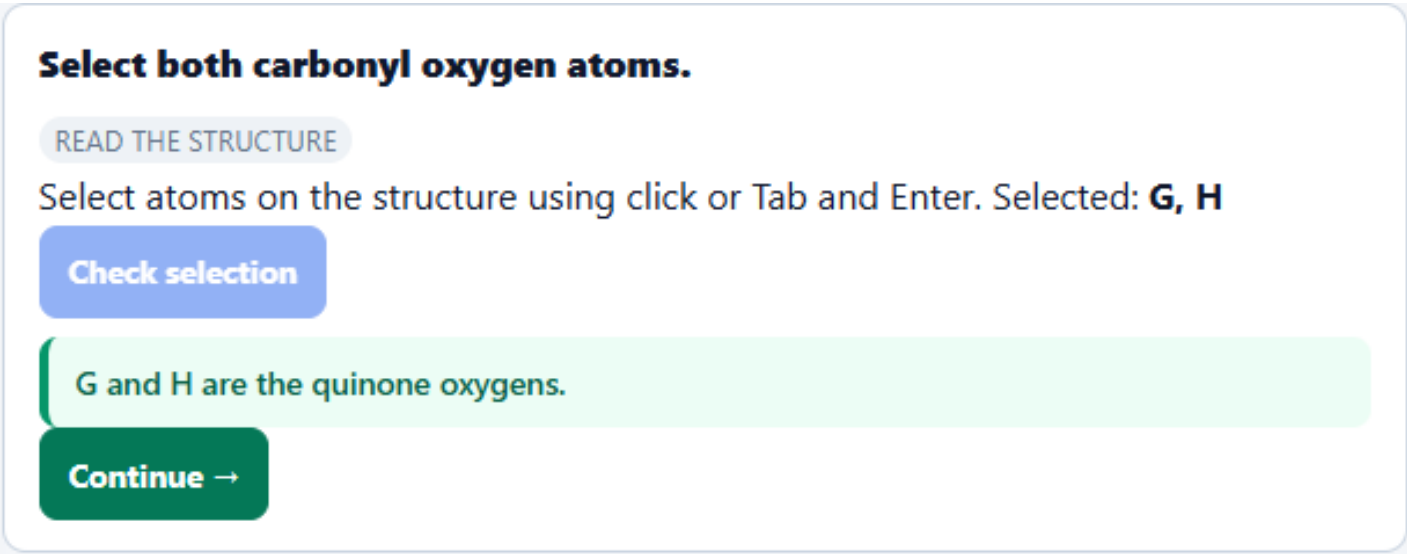
{"authorAnswer": [6, 7], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [6, 7]}}

Hint 1: Identify the two C=O bonds.

Hint 2: Select the oxygen ends.

Explanation: G and H are the quinone oxygens.

Screenshot: quinone-redox-stage-1-feedback



quinone-redox:stage-2 | Quinone → hydroquinone is…

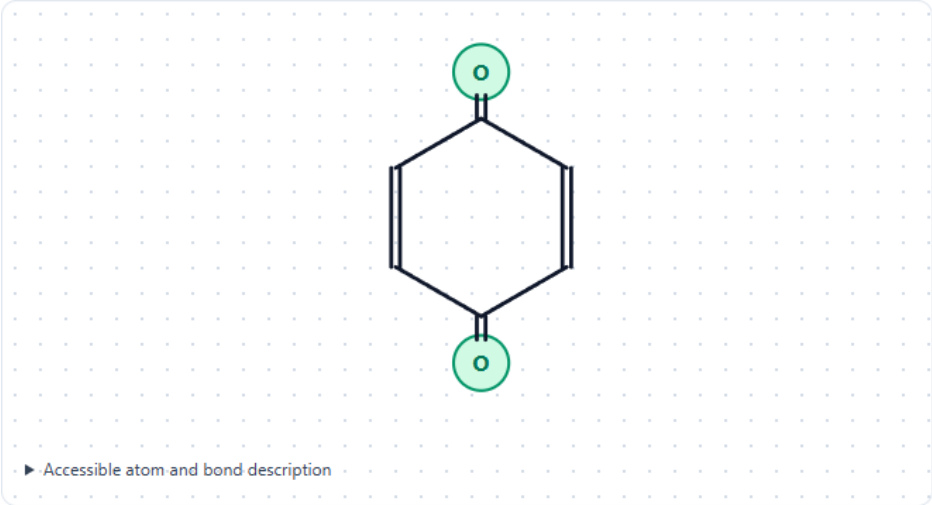
Answer type: choice

Option 0 (A): Reduction

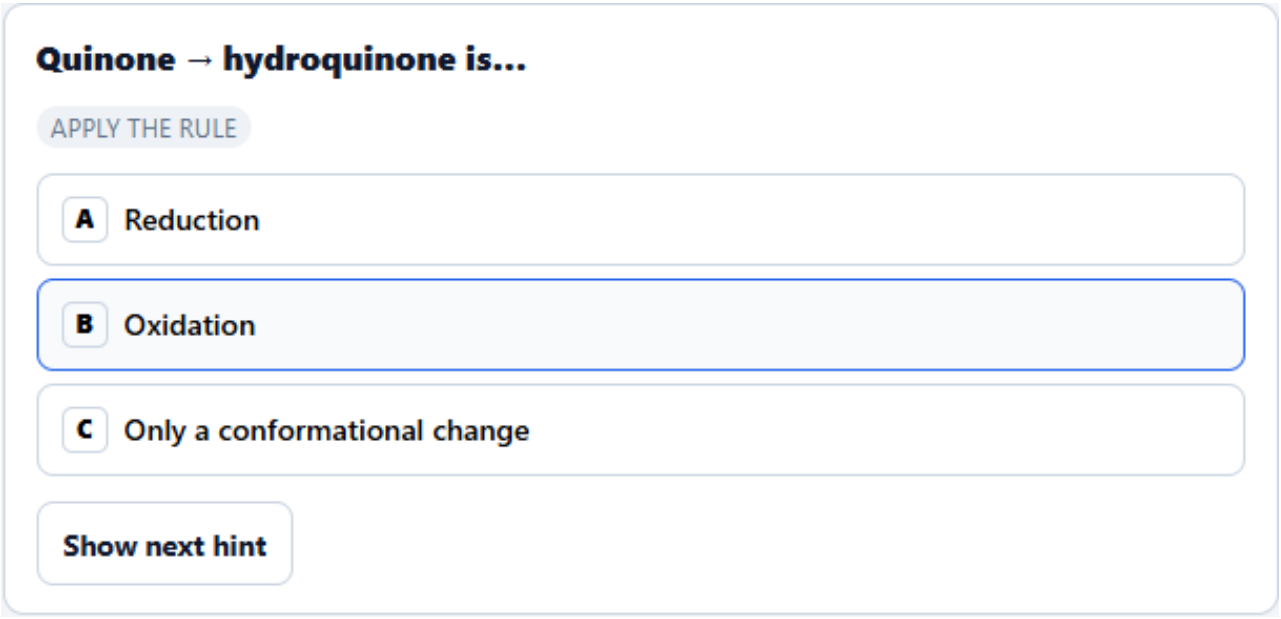
Option 1 (B): Oxidation

Option 2 (C): Only a conformational change

Screenshot: quinone-redox-stage-2-molecule



Screenshot: quinone-redox-stage-2-question



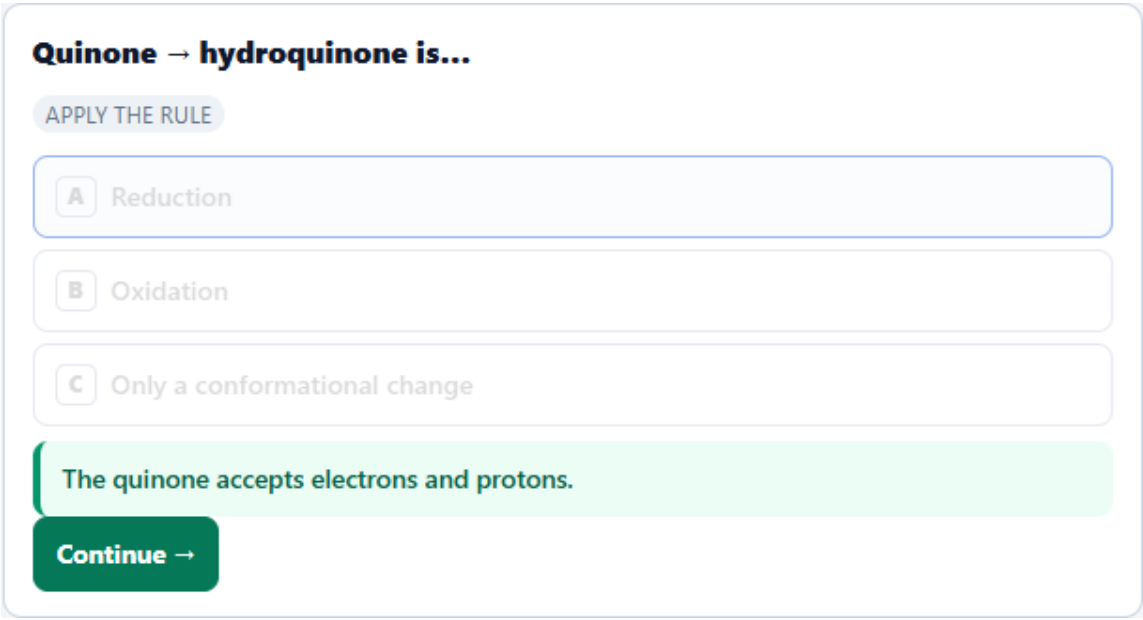
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Track C=O versus C–OH.

Hint 2: The reduced form carries more hydrogen.

Explanation: The quinone accepts electrons and protons.

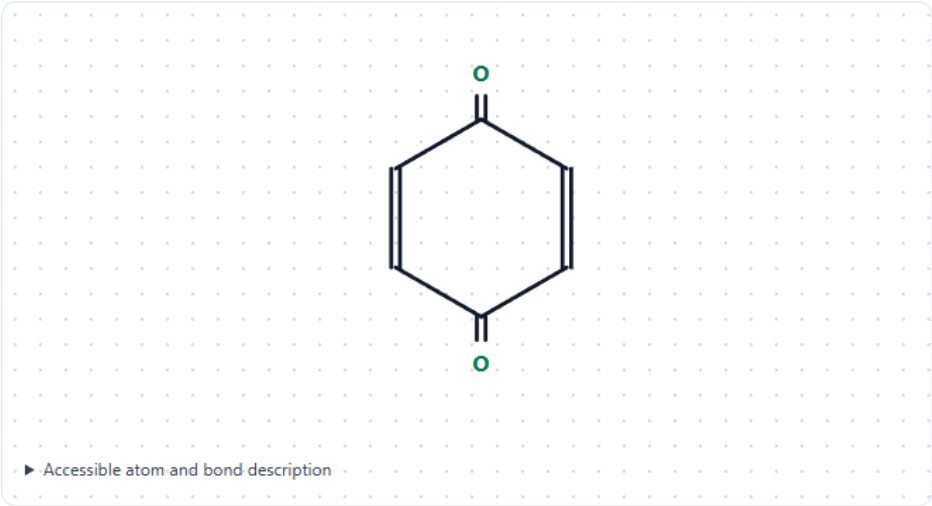
Screenshot: quinone-redox-stage-2-feedback



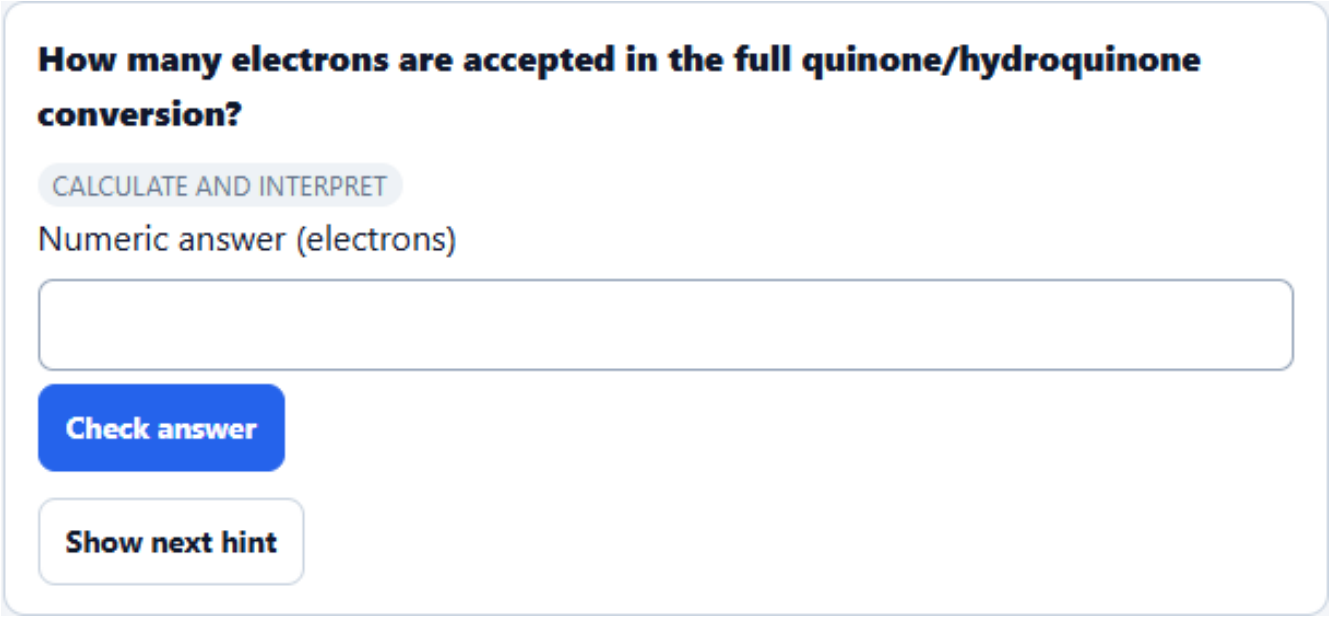
quinone-redox:stage-3 | How many electrons are accepted in the full quinone/hydroquinone conversion?

Answer type: number | unit: electrons

Screenshot: quinone-redox-stage-3-molecule



Screenshot: quinone-redox-stage-3-question



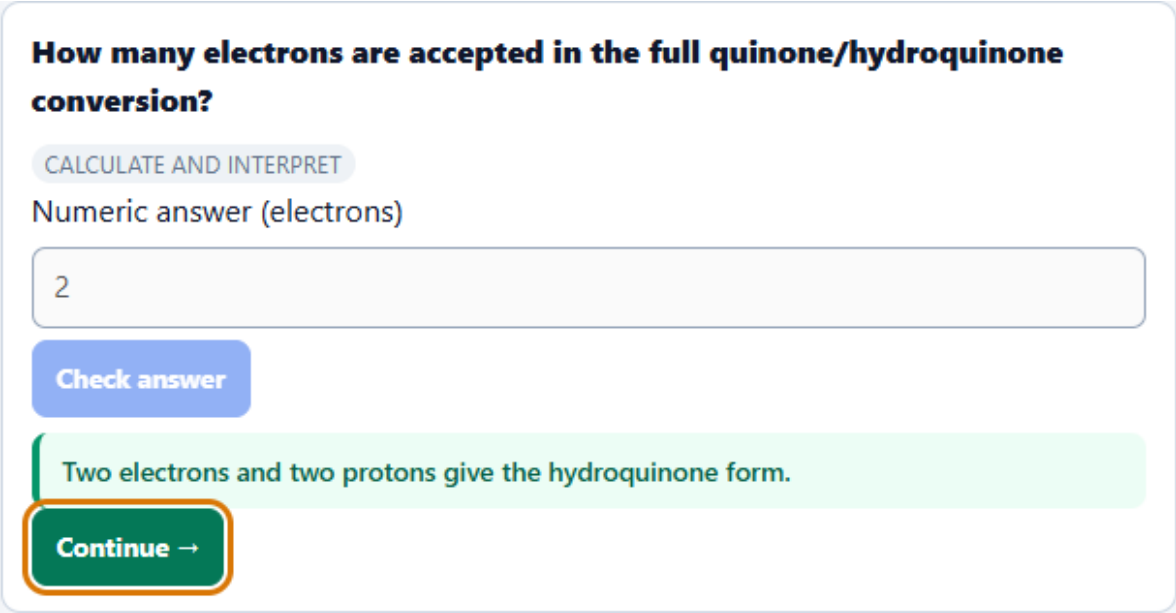
{"authorAnswer": 2, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: There are two carbonyl oxygens converted to OH.

Hint 2: Each coupled reduction contributes one electron in the overall pair.

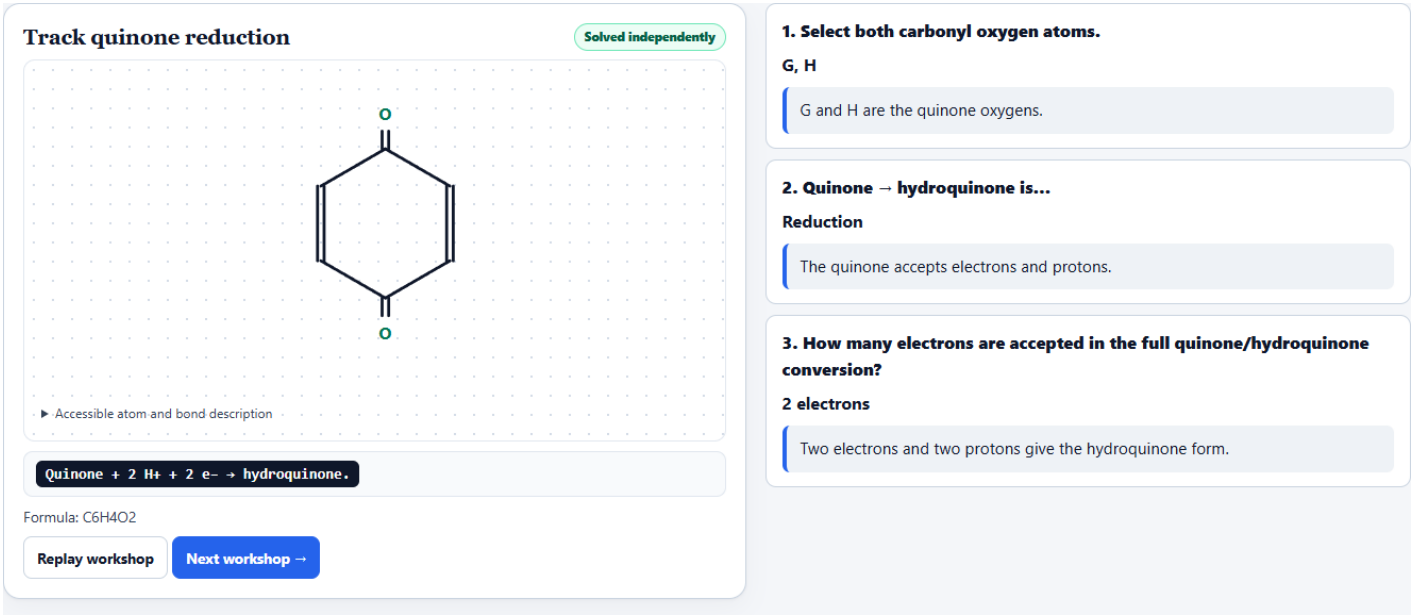
Explanation: Two electrons and two protons give the hydroquinone form.

Screenshot: quinone-redox-stage-3-feedback



Completion summary: Quinone + 2 H+ + 2 e− → hydroquinone.

Screenshot: quinone-redox-complete



Molecular repository: author claims to verify

isopropanol

Author claims: propan-2-ol | formula C3H8O

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1]]

tbutanol

Author claims: 2-methylpropan-2-ol | formula C4H10O

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=108.00, y=25.00; charge=0; explicit H override=none

4 (E): O; x=108.00, y=160.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [1, 4, 1]]

acetone

Author claims: propanone | formula C3H6O

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 2]]

acetaldehyde

Author claims: ethanal | formula C2H4O

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2]]

mesylate

Author claims: ethyl methanesulfonate | formula C3H8O3S

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=35.00, y=130.00; charge=0; explicit H override=none

1 (B): C; x=85.00, y=100.00; charge=0; explicit H override=none

2 (C): O; x=140.00, y=130.00; charge=0; explicit H override=none

3 (D): S; x=200.00, y=100.00; charge=0; explicit H override=none

4 (E): C; x=255.00, y=130.00; charge=0; explicit H override=none

5 (F): O; x=200.00, y=35.00; charge=0; explicit H override=none

6 (G): O; x=255.00, y=60.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [3, 5, 2], [3, 6, 2]]

acetal

Author claims: 2,2-dimethoxypropane | formula C5H12O2

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=75.00, y=130.00; charge=0; explicit H override=none

1 (B): C; x=135.00, y=100.00; charge=0; explicit H override=none

2 (C): C; x=195.00, y=130.00; charge=0; explicit H override=none

3 (D): O; x=105.00, y=45.00; charge=0; explicit H override=none

4 (E): O; x=175.00, y=45.00; charge=0; explicit H override=none

5 (F): C; x=50.00, y=15.00; charge=0; explicit H override=none

6 (G): C; x=230.00, y=15.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [1, 4, 1], [3, 5, 1], [4, 6, 1]]

phenol

Author claims: phenol | formula C6H6O

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=170.00, y=37.00; charge=0; explicit H override=none

1 (B): C; x=228.89, y=71.00; charge=0; explicit H override=none

2 (C): C; x=228.89, y=139.00; charge=0; explicit H override=none

3 (D): C; x=170.00, y=173.00; charge=0; explicit H override=none

4 (E): C; x=111.11, y=139.00; charge=0; explicit H override=none

5 (F): C; x=111.11, y=71.00; charge=0; explicit H override=none

6 (G): O; x=170.00, y=5.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 1]]

hydroquinone

Author claims: benzene-1,4-diol (hydroquinone) | formula C6H6O2

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=170.00, y=37.00; charge=0; explicit H override=none

1 (B): C; x=228.89, y=71.00; charge=0; explicit H override=none

2 (C): C; x=228.89, y=139.00; charge=0; explicit H override=none

3 (D): C; x=170.00, y=173.00; charge=0; explicit H override=none

4 (E): C; x=111.11, y=139.00; charge=0; explicit H override=none

5 (F): C; x=111.11, y=71.00; charge=0; explicit H override=none

6 (G): O; x=170.00, y=5.00; charge=0; explicit H override=none

7 (H): O; x=170.00, y=205.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 1], [3, 7, 1]]

quinone

Author claims: 1,4-benzoquinone | formula C6H4O2

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=170.00, y=37.00; charge=0; explicit H override=none

1 (B): C; x=228.89, y=71.00; charge=0; explicit H override=none

2 (C): C; x=228.89, y=139.00; charge=0; explicit H override=none

3 (D): C; x=170.00, y=173.00; charge=0; explicit H override=none

4 (E): C; x=111.11, y=139.00; charge=0; explicit H override=none

5 (F): C; x=111.11, y=71.00; charge=0; explicit H override=none

6 (G): O; x=170.00, y=5.00; charge=0; explicit H override=none

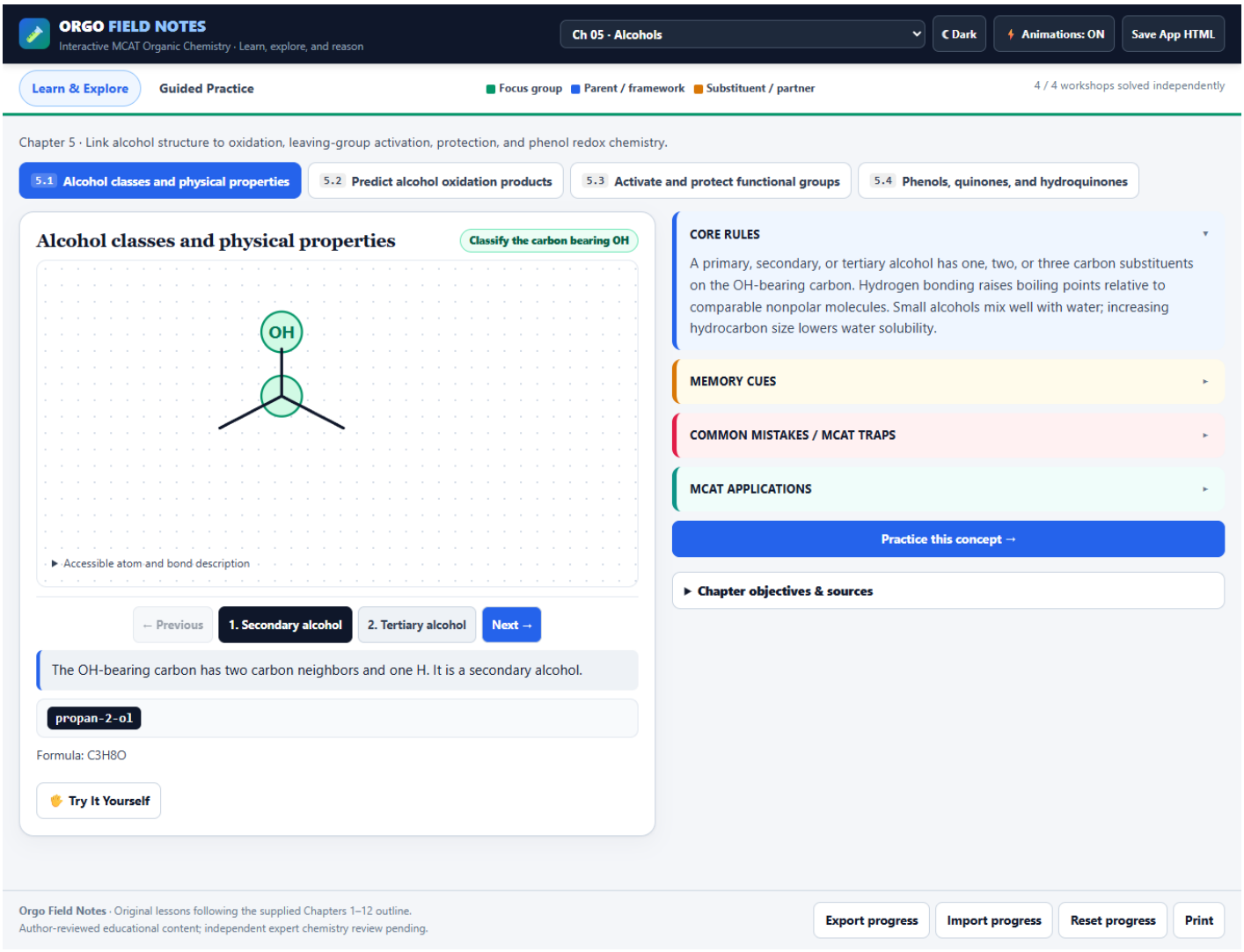
7 (H): O; x=170.00, y=205.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [2, 3, 1], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 2], [3, 7, 2]]

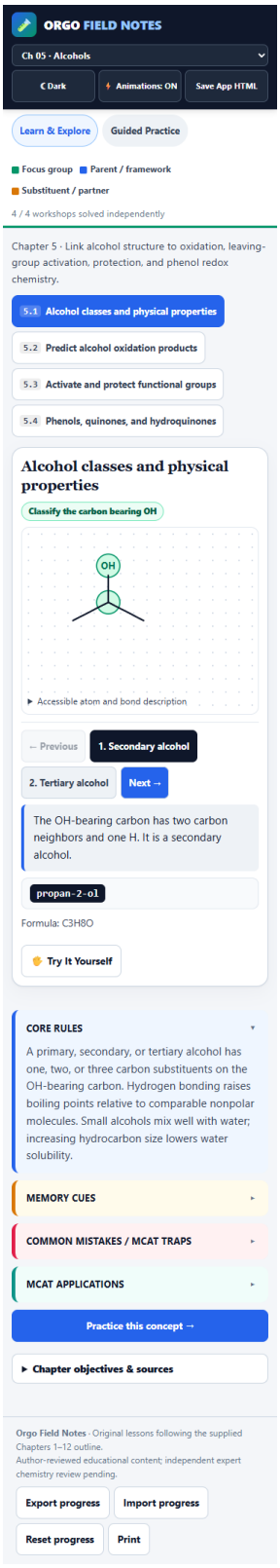
Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

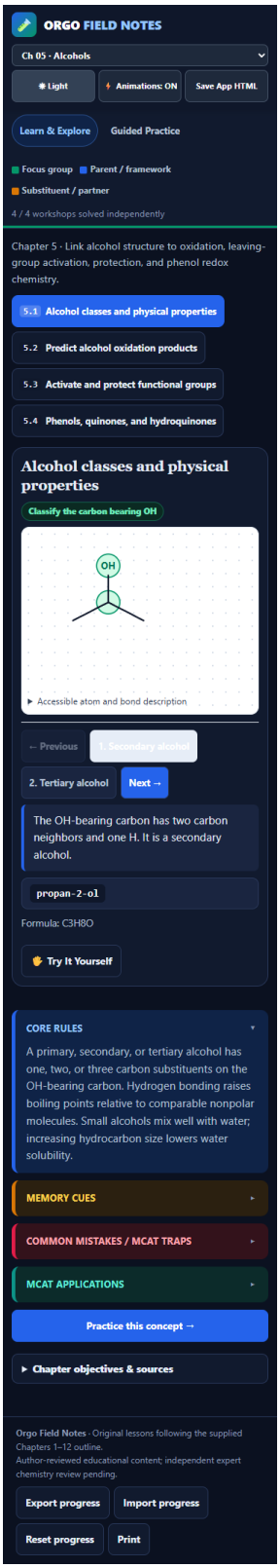
Screenshot: layout-desktop-light



Screenshot: layout-mobile-light



Screenshot: layout-mobile-dark



Execution evidence

55 screenshots and 13 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.